

BioFlow AI Scientist Integrated Scientific Report

Publication-style scientific report generated from the current BioFlow analysis session.

Portfolio edition: internal project/session identifiers, dataset record IDs, cryptographic hashes, runtime bundle filenames, local/internal paths, and sample/subject identifiers have been omitted or anonymized. Scientific methods, parameters, figures, numerical results, statistical boundaries, and reproducibility software versions are unchanged.

Expression matrix	TCGA_BRCA_ER_log2TPM1_expression.csv (22184 × 779)	Metadata	TCGA_BRCA_ER_metadata.csv (778 × 5)
Genes	22184	Samples	778

1. Cohort Overview

This report integrates quality control, feature discovery, predictive modeling, validation analysis, perturbation simulation, and AI Scientist reasoning outputs for hypothesis generation and downstream validation planning.

Genes	Samples	AUC	Target score
22184	778	0.970	78

2. Quality Control

Figure 1. PCA / embedding sample-space plot

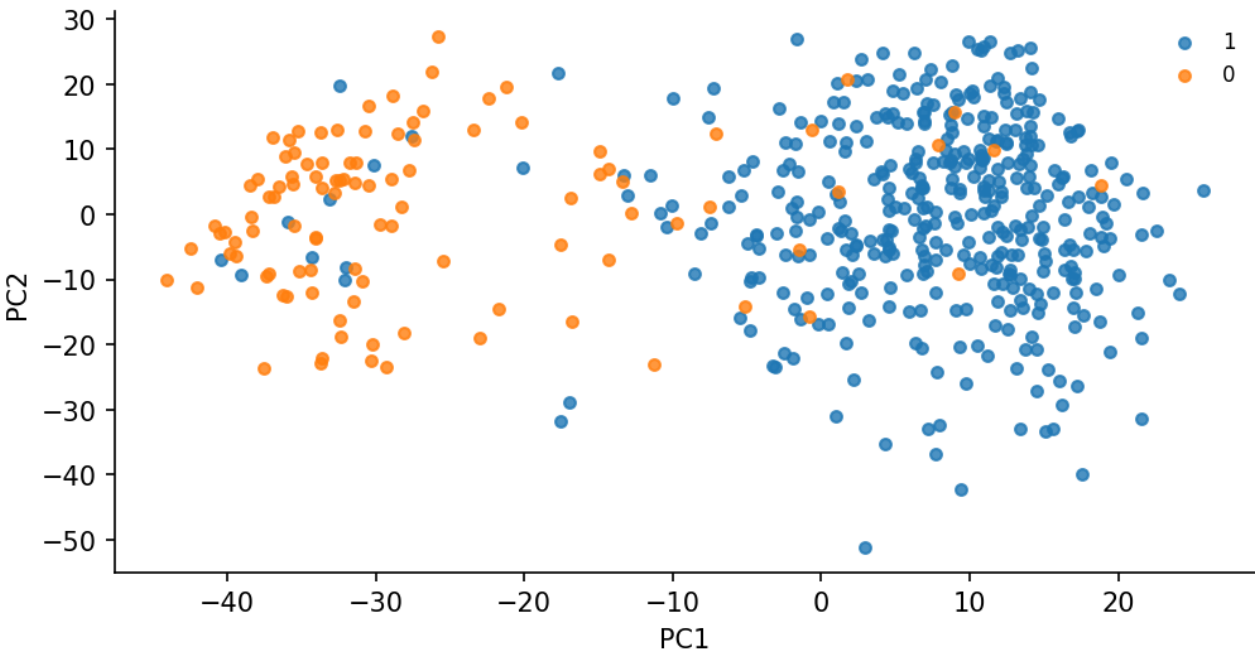
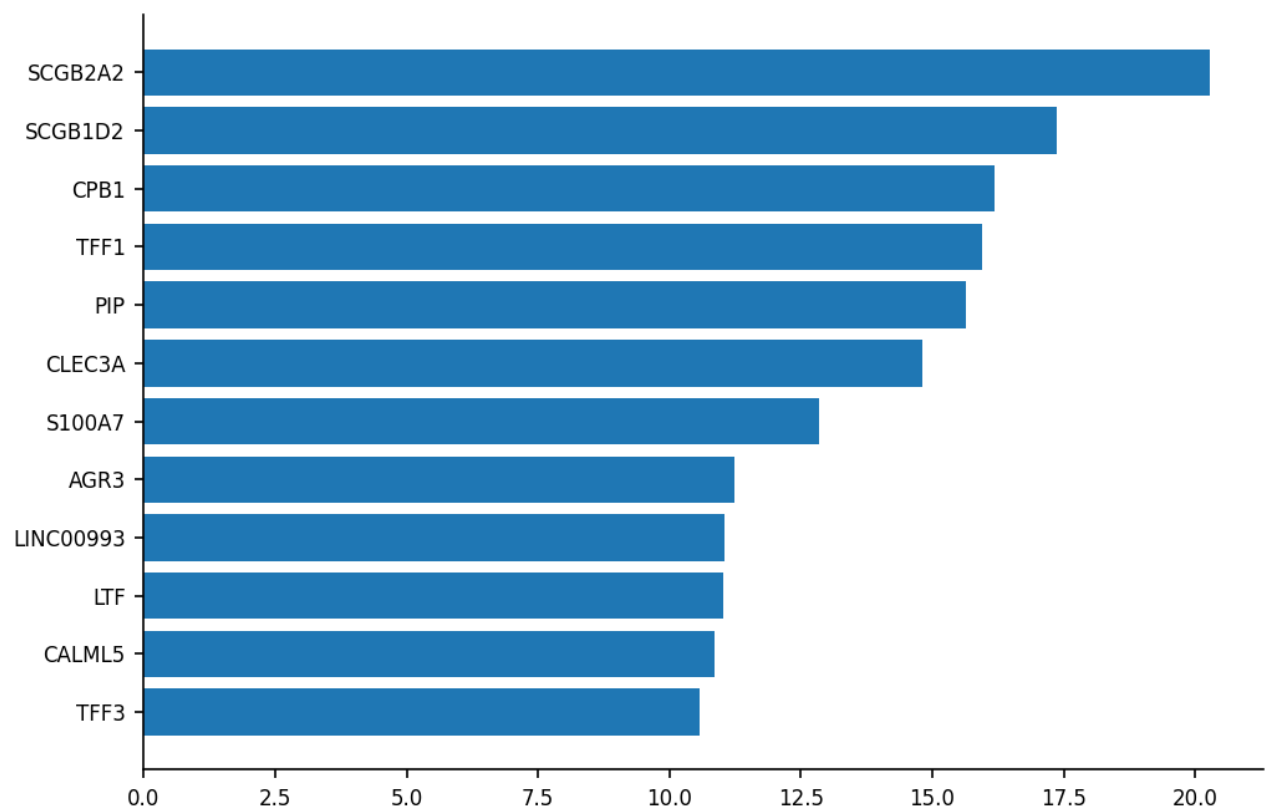


Figure 1. Sample distribution on the first two components for grouping, batch-effect, and outlier inspection.

Figure 2. Top variable genes



3. Feature Discovery

This section reports the feature-screening method actually executed in this Analysis, its configuration, candidate genes, and fold-level stability. Screening methods that were not executed are not presented as comparative evidence.

Table 1A. Feature-screening method and configuration

Method	Algorithm ID	Version	Candidate genes	Top N	Variance threshold	Seed	Selection scope
Welch delta score	feature.welch_delta_score	1.0.0	50	50	0	7	training_subset_only

Table 1A-2. Complete Feature Selection parameters

Parameter	Value
algorithm	feature.welch_delta_score
algorithm_version	1.0.0
analysis_speed	full
max_features	2000
random_seed	7
requested_algorithm	auto
test_size	0.2
top_n	50
validation_strategy	internal
variance_threshold	0

Table 1A-3. Complete Feature Selection protocol

Protocol field	Value
feature_selection_scope	training_subset_only

holdout_samples	156
leakage_prevention	True
mode	internal_holdout_mode
random_seed	7
test_size	0.2
train_samples	622
validation_strategy	internal

Table 1B. Fold-level candidate-feature stability

Gene	Selection frequency	Selected folds	Total folds
ESR1	1.000	9	9
AGR3	1.000	9	9
TFF1	1.000	9	9
CYP2B7P	1.000	9	9
NAT1	1.000	9	9
SCUBE2	1.000	9	9
FSIP1	1.000	9	9
CCDC170	1.000	9	9
GFRA1	1.000	9	9
POTEKP	1.000	9	9
CA12	1.000	9	9
AFF3	1.000	9	9
SERPINA11	1.000	9	9
SLC39A6	1.000	9	9
C5AR2	1.000	9	9
TFF3	1.000	9	9
LINC01087	1.000	9	9
THSD4	1.000	9	9
TBC1D9	1.000	9	9
SLC7A2	1.000	9	9
DACH1	1.000	9	9
SLC16A6	1.000	9	9
DNAL1	1.000	9	9
GREB1	1.000	9	9
GATA3	1.000	9	9
GATA3-AS1	1.000	9	9
CLSTN2	1.000	9	9
MAPT	1.000	9	9
PGR	1.000	9	9
ABAT	1.000	9	9
ZMYND10	1.000	9	9
DEGS2	1.000	9	9
AGR2	1.000	9	9
ERBB4	1.000	9	9
LINC00993	1.000	9	9
KCNJ3	0.889	8	9
TPRG1	0.889	8	9
CT62	0.889	8	9
SUSD3	0.889	8	9
PSAT1	0.889	8	9
TNRC18P1	0.889	8	9
DNAJC12	0.778	7	9
RGS22	0.667	6	9
KCNE4	0.667	6	9
ADAMTS15	0.667	6	9
KCNK15	0.667	6	9
MLPH	0.556	5	9
AKR7A3	0.556	5	9
GRPR	0.556	5	9
ANKRD30A	0.556	5	9
CHAD	0.444	4	9
FOXA1	0.444	4	9
SLC44A4	0.333	3	9
STC2	0.333	3	9
BMPR1B	0.333	3	9
GP2	0.333	3	9
CHRD	0.333	3	9
STMND1	0.333	3	9
CST9	0.222	2	9
CPB1	0.222	2	9
SYT9	0.222	2	9
PRR15	0.111	1	9
EVL	0.111	1	9
WNK4	0.111	1	9
IL20	0.111	1	9

Table 1C. All saved ranked Feature Selection results

Gene	Score	Delta	P Value	Method
ESR1	454.32577219139324	-5.263165581222064	4.766826562508772e-87	Welch delta score
AGR3	448.50372370341876	-6.194983128759891	4.0004359472668875e-73	Welch delta score
TFF1	367.21806095305607	-6.174580123986148	3.368559338224473e-60	Welch delta score
CYP2B7P	347.18174594964466	-4.463817961779704	1.67162830131085e-78	Welch delta score
NAT1	302.7949555162608	-3.9142216420128246	4.388925906156023e-78	Welch delta score
SCUBE2	284.3319176847505	-4.090632754553672	3.1041818274368354e-70	Welch delta score
POTEKP	264.33007830788193	-2.7403027999774605	3.46599877651889e-97	Welch delta score
GFRA1	238.84306520383018	-4.325417286962231	6.046408932207687e-56	Welch delta score
CCDC170	235.72319443003704	-2.853034645844384	2.388262423569718e-83	Welch delta score
LINC01087	235.15651542906204	-2.576905956635619	5.5542643510767186e-92	Welch delta score
CA12	232.2632069458501	-4.135173672638004	6.796681832080048e-57	Welch delta score
SERPINA11	227.10660237009935	-3.415039629997535	3.1485190474197096e-67	Welch delta score
FSIP1	213.50945365796778	-3.0046773097501998	8.72912028272007e-72	Welch delta score
C5AR2	210.65901648149963	-2.529868852219252	5.385799731075388e-84	Welch delta score
DACH1	206.35025752095135	-2.921517333229215	2.3377927406658433e-71	Welch delta score
DNALI1	203.15975818826885	-3.3482452852376525	2.1062697757625603e-61	Welch delta score
SLC39A6	199.78111867938065	-2.7997105551746495	4.387569344303038e-72	Welch delta score
THSD4	199.16801243471488	-2.983151652634297	1.7207053896960772e-67	Welch delta score
TFF3	199.09752533163052	-4.933847143297772	4.431965263178944e-41	Welch delta score
SLC7A2	196.7324329931987	-3.598864988031472	2.1620566193914302e-55	Welch delta score
TBC1D9	194.68279653529018	-3.2357147258602916	6.809880609386143e-61	Welch delta score
CLSTN2	189.7770443027208	-3.3017449483222885	3.3281080833409984e-58	Welch delta score
GATA3-AS1	182.82000201752774	-2.56872376441585	6.737044158531136e-72	Welch delta score
SLC16A6	180.36928411631203	-2.6798167675324276	4.936584128509931e-68	Welch delta score
AGR2	180.332007933456	-4.7662054485794005	1.4603242334985447e-38	Welch delta score
AFF3	172.61927129541507	-3.120356041596361	4.782152447158245e-56	Welch delta score
PGR	169.8284811028285	-3.160696787346841	1.8563406011523737e-54	Welch delta score
MAPT	169.19700873124344	-2.9627564742662775	7.798834468321147e-58	Welch delta score
ERBB4	167.83621606205267	-2.7570816365182402	1.3347788968027216e-61	Welch delta score
GREB1	167.7098959964742	-2.9729123639013384	3.866690943066485e-57	Welch delta score
GATA3	167.46119662362702	-3.5255679347560394	3.1690597330634004e-48	Welch delta score
CT62	167.4074105849185	-2.482668156923303	3.711574924075235e-68	Welch delta score
SUSD3	161.61456377314929	-3.00689478857157	1.786511273486852e-54	Welch delta score
TNRC18P1	160.41556424117985	-2.6217001088719627	6.492169218455505e-62	Welch delta score
LINC00993	157.60119269218532	-3.9869309669110935	2.9549415913726768e-40	Welch delta score
TPRG1	156.70672781412765	-2.441566841666236	6.563660800767997e-65	Welch delta score
ZMYND10	152.92467491410568	-2.400293247278875	1.946121933148472e-64	Welch delta score
DNAJC12	152.57934068759727	-3.0208487896433285	3.0990979841458477e-51	Welch delta score
ABAT	152.03710796155258	-2.471681408379923	3.0788457158237114e-62	Welch delta score
KCNJ3	150.6712642083876	-2.778339477546578	5.879101749164928e-55	Welch delta score
PSAT1	149.8112728404827	3.2227299091779176	3.267161292254611e-47	Welch delta score
DEGS2	148.34219948399294	-3.12548681039388	3.450561566281644e-48	Welch delta score
GRPR	148.29913049511103	-2.5925631893373393	6.284316788765027e-58	Welch delta score
ADAMTS15	142.78862563033516	-2.7818407813305055	4.689996171041867e-52	Welch delta score
KCNE4	142.6023181259427	-2.4690079665193525	1.7501322881895726e-58	Welch delta score
ANKRD30A	140.6926566540331	-3.1710506683289506	4.287101775788383e-45	Welch delta score
MLPH	139.7994585377157	-3.472160863925937	5.45813718229065e-41	Welch delta score
RGS22	138.9984575980513	-1.6805150283067287	1.9416876289950713e-83	Welch delta score
FOXA1	138.43461302564765	-3.976359478200882	1.5331654832157156e-35	Welch delta score
SLC44A4	137.28398452544442	-3.828142666452516	1.3747593562187545e-36	Welch delta score

Figure 3. Feature significance plot

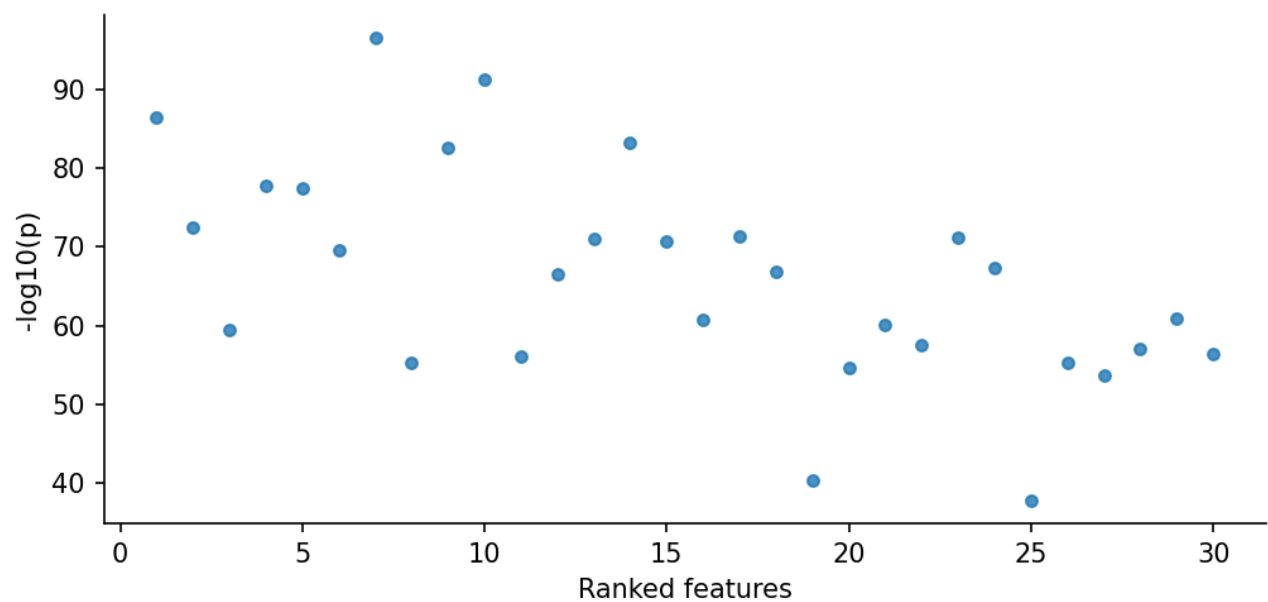
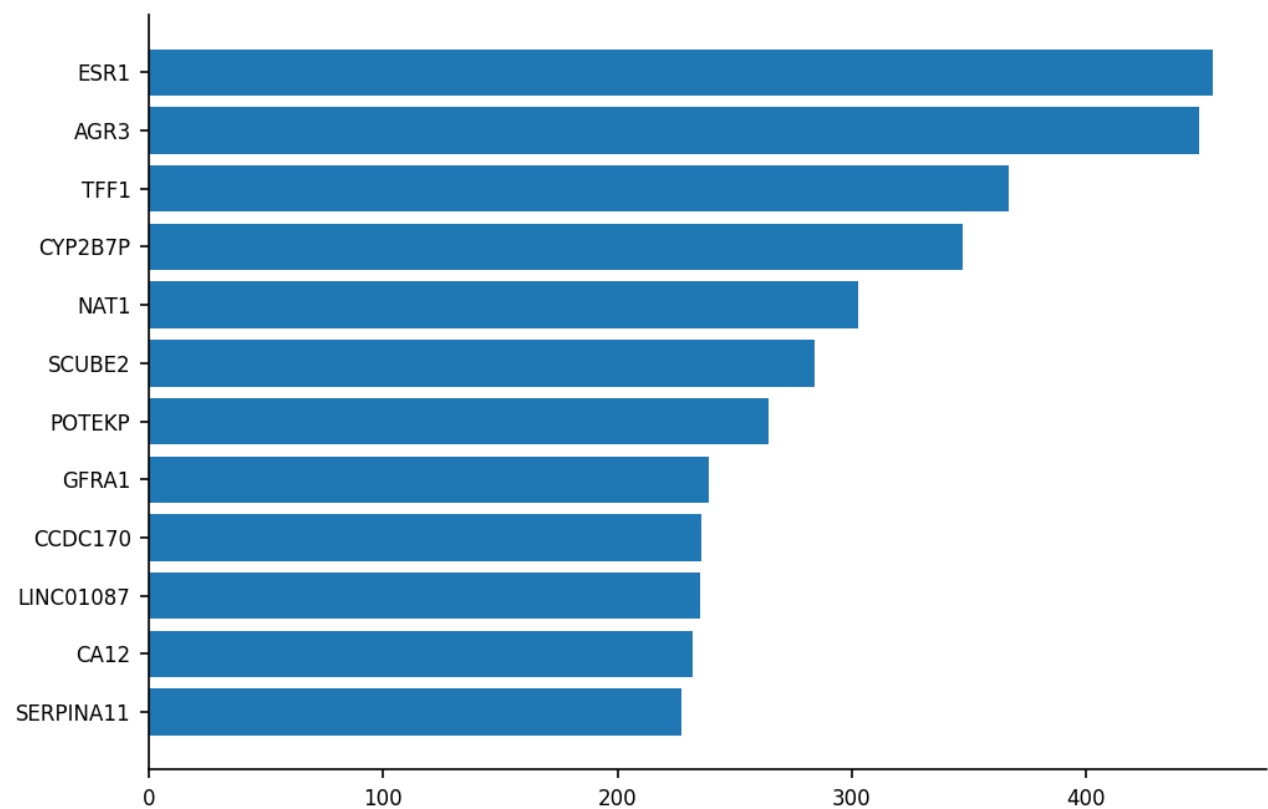


Figure 4. Feature ranking



4. Predictive Modeling

Selected model: Logistic Regression. Model selection was completed from repeated-CV performance before evaluation on the untouched outer holdout; the outer holdout was not used to choose among candidate models.

Complete Modeling run parameters

Parameter	Value
algorithm_selection_mode	user_selected
algorithms	model.logistic_regression, model.random_forest
analysis_speed	quick
candidate_feature_count	50
cv_folds	3
cv_repeats	3
max_features	1000
random_seed	7
test_size	0.2
top_k	30
tune	False
validation_strategy	internal
variance_threshold	0

Complete Modeling protocol

Protocol field	Value
algorithm_selection_mode	user_selected
candidate_pool_source	saved_feature_selection
cv_folds	3
cv_repeats	3
evaluation_scope	repeated_nested_cv_on_outer_train_then_single_untouched_holdout
feature_screening_inside_cv	True
feature_selection_inside_cv	True
holdout_used_for_model_selection	False
leakage_note	V5 uses two fold-local selection layers: Stage-1 feature screening creates a candidate pool from fold-training samples, then each Modeling Adapter independently re-ranks/re-selects within that pool using the same fold-training samples. Candidate models are selected by repeated CV performance. The untouched outer holdout is evaluated only after the winner is fixed.
mode	internal_holdout_mode
model_selection_basis	nested_cv_auc_mean
model_specific_selection_inside_cv	True
n_test	156
n_train	622
outer_holdout_sample_ids	156 holdout samples (identifiers omitted in portfolio version)
outer_train_sample_ids	622 training samples (identifiers omitted in portfolio version)
random_seed	7
selected_model_algorithms	model.logistic_regression, model.random_forest
test_size	0.2
training_scope	outer_train_subset_with_untouched_holdout
tuning_applied	False
tuning_requested	False
validation_strategy	internal

Nested / repeated CV configuration

Field	Value
available	True
candidate_top_n_per_fold	50
effective_folds	3
feature_selection_algorithm	feature.welch_delta_score
leakage_prevention	Every CV fold reruns Stage-1 feature screening on fold-training samples only. Each Modeling Adapter then performs its own Stage-2 re-ranking/selection inside that candidate pool using the same fold-training samples. CV validation samples participate only in evaluation.
mean_pairwise_feature_jaccard	0.7851214135727155
mode	repeated_stratified_nested_feature_selection
model_top_k_per_fold	30
negative_class	0

oof_prediction_scope	repeated_cv_validation_folds_within_outer_training
out_of_sample	True
positive_class	1
random_seed	7
repeats	3
requested_folds	3
runtime	algorithm_registry_adapters
selection_architecture	two_stage_screening_then_model_specific_selection
successful_feature_selection_folds	9
total_folds	9

Table 2A. Modeling methods and repeated-CV comparison

Role	Model	Algorithm ID	Version	CV rank	CV AUC	AUC SD	CV accuracy	Selected genes	Fold Jaccard	Selection method
Selected	Logistic Regression	model.logistic_regression	1.0.0	1	0.964	0.015	0.929	30	0.447	embedded_l1_ranking_then_l2_logistic_fit
Comparison	Random Forest	model.random_forest	1.0.0	2	0.953	0.021	0.943	30	0.565	random_forest_candidate_ranking_then_refit

4.1 Candidate-model repeated-CV out-of-fold diagnostics

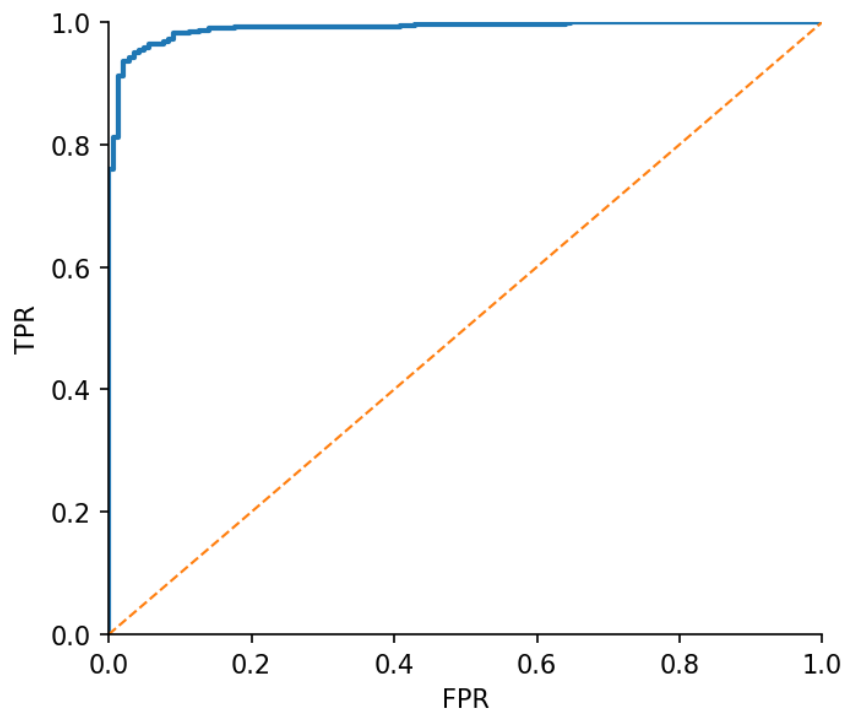
The ROC curves and confusion matrices below use repeated-CV out-of-fold predictions within the outer-training cohort so every candidate model can be compared on the same leakage-safe basis. They are not training-resubstitution performance and do not use the untouched outer holdout.

Logistic Regression · Selected

Logistic Regression · apparent outer-training fit

ROC AUC	PR-AUC	Accuracy	Balanced accuracy	Sensitivity	Specificity	Reference threshold	Training N
0.989	0.997	0.952	0.956	0.948	0.965	0.500	622

Logistic Regression · apparent training ROC



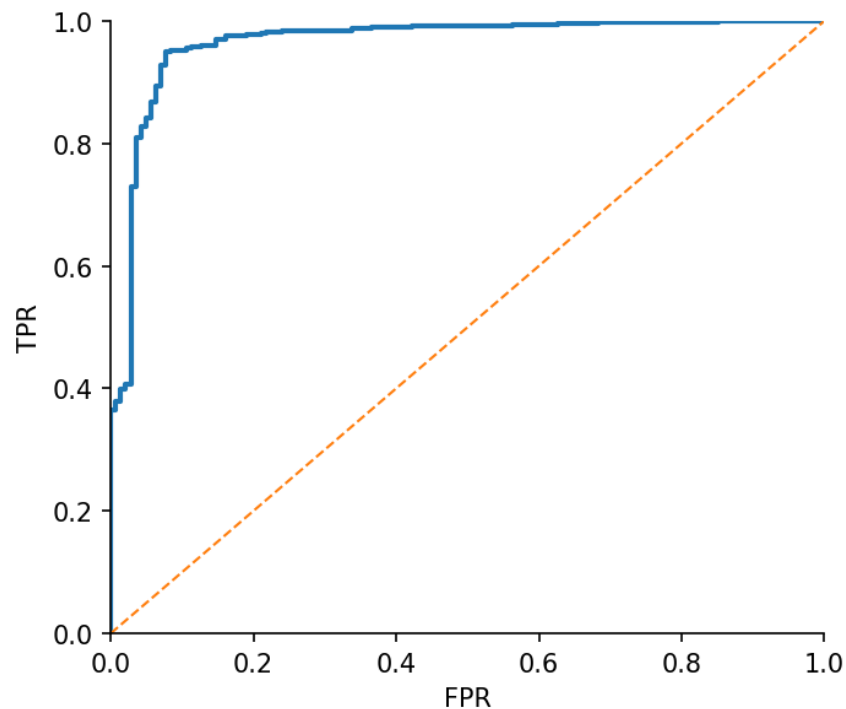
Logistic Regression · apparent training confusion matrix

Actual class	Pred 0	Pred 1
Actual 0	137	5
Actual 1	25	455

This is apparent/resubstitution performance on the same outer-training samples used to fit the model. It is expected to be optimistic, is shown for transparency, and is not validation evidence.

ROC AUC	PR-AUC	Accuracy	Balanced accuracy	Sensitivity	Specificity	Reference threshold	OOF N
0.965	0.987	0.939	0.933	0.944	0.923	0.500	622

Logistic Regression · repeated-CV OOF ROC



Logistic Regression · repeated-CV OOF confusion matrix

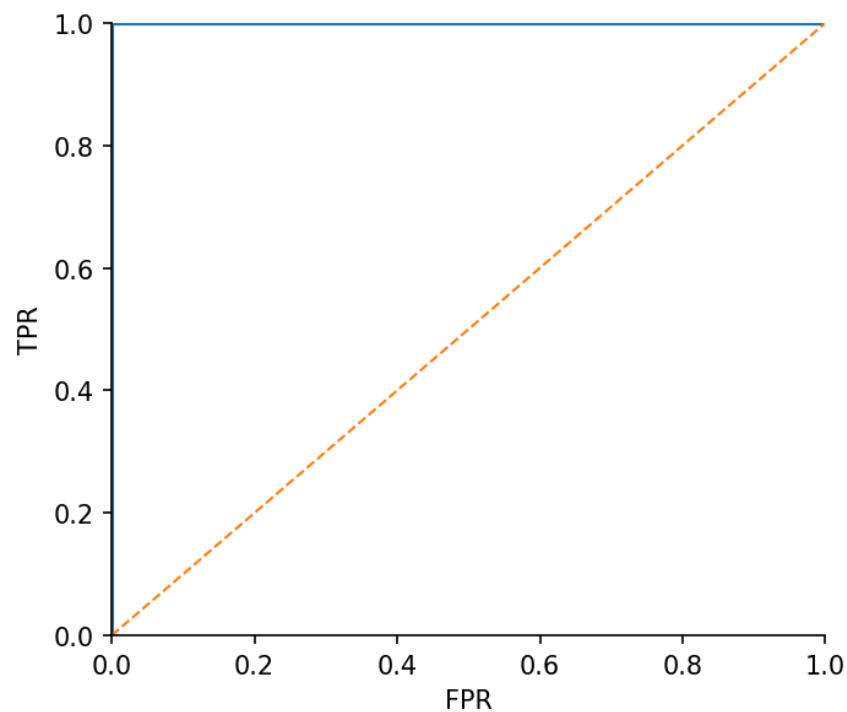
Actual class	Pred 0	Pred 1
Actual 0	131	11
Actual 1	27	453

Random Forest · Comparison

Random Forest · apparent outer-training fit

ROC AUC	PR-AUC	Accuracy	Balanced accuracy	Sensitivity	Specificity	Reference threshold	Training N
1.000	1.000	1.000	1.000	1.000	1.000	0.500	622

Random Forest · apparent training ROC



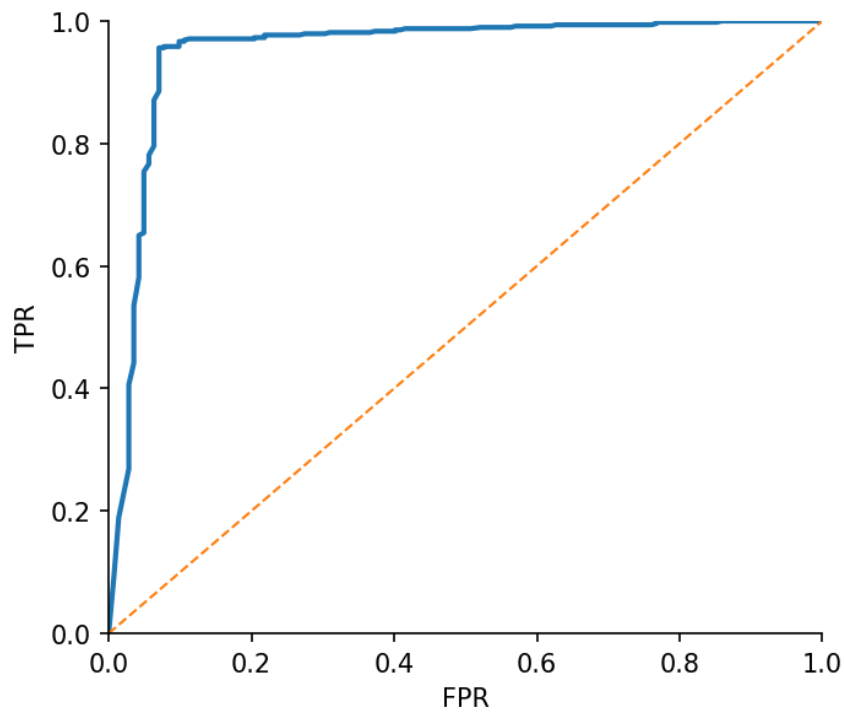
Random Forest · apparent training confusion matrix

Actual class	Pred 0	Pred 1
Actual 0	142	0
Actual 1	0	480

This is apparent/resubstitution performance on the same outer-training samples used to fit the model. It is expected to be optimistic, is shown for transparency, and is not validation evidence.

ROC AUC	PR-AUC	Accuracy	Balanced accuracy	Sensitivity	Specificity	Reference threshold	OOF N
0.951	0.975	0.952	0.929	0.971	0.887	0.500	622

Random Forest · repeated-CV OOF ROC



Random Forest · repeated-CV OOF confusion matrix

Actual class	Pred 0	Pred 1
Actual 0	126	16
Actual 1	14	466

Table 2B. Model-specific selected genes, importance, and fold stability

Model	Rank	Gene	Importance	Selection frequency	Selected folds	Total folds
Logistic Regression	1	ESR1	1.716	1.000	9	9
Logistic Regression	2	TBC1D9	1.736	1.000	9	9
Logistic Regression	3	FOXA1	1.308	0.333	3	9
Logistic Regression	4	PGR	1.118	1.000	9	9
Logistic Regression	5	DACH1	1.013	1.000	9	9
Logistic Regression	6	CYP2B7P	0.835	1.000	9	9
Logistic Regression	7	POTEKP	0.806	0.889	8	9
Logistic Regression	8	SLC44A4	0.671	0.333	3	9
Logistic Regression	9	CA12	0.682	0.889	8	9
Logistic Regression	10	DNAL1	0.609	0.889	8	9
Logistic Regression	11	LINC00993	0.475	0.889	8	9
Logistic Regression	12	RGS22	0.458	0.667	6	9
Logistic Regression	13	ABAT	0.457	0.667	6	9
Logistic Regression	14	GATA3	0.567	0.778	7	9
Logistic Regression	15	SLC7A2	0.363	0.778	7	9
Logistic Regression	16	AGR2	0.438	0.778	7	9
Logistic Regression	17	ERBB4	0.364	0.333	3	9
Logistic Regression	18	C5AR2	0.547	0.333	3	9
Logistic Regression	19	AGR3	0.441	0.889	8	9
Logistic Regression	20	DNAJC12	0.476	0.667	6	9
Logistic Regression	21	DEGS2	0.421	0.667	6	9
Logistic Regression	22	GATA3-AS1	0.393	0.667	6	9
Logistic Regression	23	SLC16A6	0.295	0.667	6	9
Logistic Regression	24	CT62	0.398	0.333	3	9
Logistic Regression	25	TNRC18P1	0.281	0.778	7	9
Logistic Regression	26	ADAMTS15	0.328	0.444	4	9
Logistic Regression	27	SLC39A6	0.250	0.333	3	9
Logistic Regression	28	KCNJ3	0.303	0.667	6	9
Logistic Regression	29	GREB1	0.312	0.444	4	9
Logistic Regression	30	TFF3	0.292	—	—	—
Random Forest	1	ESR1	0.172	1.000	9	9
Random Forest	2	AGR3	0.098	1.000	9	9
Random Forest	3	CCDC170	0.117	1.000	9	9
Random Forest	4	GATA3	0.071	1.000	9	9
Random Forest	5	C5AR2	0.107	1.000	9	9
Random Forest	6	TBC1D9	0.045	1.000	9	9
Random Forest	7	CT62	0.042	0.889	8	9
Random Forest	8	POTEKP	0.020	1.000	9	9
Random Forest	9	THSD4	0.041	1.000	9	9
Random Forest	10	DNAL1	0.035	0.889	8	9
Random Forest	11	LINC01087	0.018	0.889	8	9
Random Forest	12	DEGS2	0.007	0.667	6	9
Random Forest	13	NAT1	0.010	0.778	7	9
Random Forest	14	ZMYND10	0.028	0.889	8	9
Random Forest	15	CYP2B7P	0.031	0.778	7	9
Random Forest	16	CA12	0.019	0.889	8	9
Random Forest	17	SLC39A6	0.016	1.000	9	9
Random Forest	18	DACH1	0.010	0.667	6	9
Random Forest	19	MLPH	0.019	0.556	5	9
Random Forest	20	GATA3-AS1	0.007	0.889	8	9
Random Forest	21	MAPT	0.009	0.444	4	9
Random Forest	22	SERPINA11	0.007	0.556	5	9
Random Forest	23	PGR	0.011	0.556	5	9
Random Forest	24	RGS22	0.011	0.667	6	9
Random Forest	25	SLC16A6	0.006	0.222	2	9
Random Forest	26	PSAT1	0.009	0.889	8	9
Random Forest	27	AGR2	0.008	0.556	5	9
Random Forest	28	ABAT	0.014	0.778	7	9
Random Forest	29	TFF1	0.006	0.556	5	9
Random Forest	30	SLC7A2	0.008	0.333	3	9

Model selection and cross-model evidence

Logistic Regression was selected from the compared algorithms using the prespecified repeated-CV selection rule.

The selected model achieved an untouched-holdout ROC AUC of 0.970 and PR-AUC of 0.987.

Cross-model concordance identified 20 genes shared across all successfully compared model-specific selected-gene sets.

Biological convergence summarized 8 shared and 0 partially shared model-supported pathways.

Field	Recorded value
Selected model	Logistic Regression
Selected algorithm	model.logistic_regression
Modeling run ID	Omitted in portfolio version
Candidate feature count	50
Final signature size	30
Holdout used for selection	False

Complete Cross-Model and Biological-Convergence Evidence

Complete model feature importance

Role	Model	Algorithm Id	Rank	Gene	Importance
Selected	Logistic Regression	model.logistic_regression	1	TBC1D9	1.7360594782987322
Selected	Logistic Regression	model.logistic_regression	2	ESR1	1.716180138976516
Selected	Logistic Regression	model.logistic_regression	3	FOXA1	1.3075801149529187
Selected	Logistic Regression	model.logistic_regression	4	PGR	1.1182862632002943
Selected	Logistic Regression	model.logistic_regression	5	DACH1	1.0132754801621904
Selected	Logistic Regression	model.logistic_regression	6	CYP2B7P	0.8351479607574845
Selected	Logistic Regression	model.logistic_regression	7	POTEKP	0.8064835640899054
Selected	Logistic Regression	model.logistic_regression	8	CA12	0.681878720567479
Selected	Logistic Regression	model.logistic_regression	9	SLC44A4	0.6712610320012808
Selected	Logistic Regression	model.logistic_regression	10	DNAL1	0.6092972876584211
Selected	Logistic Regression	model.logistic_regression	11	GATA3	0.5665274559574516
Selected	Logistic Regression	model.logistic_regression	12	C5AR2	0.5469527742078751
Selected	Logistic Regression	model.logistic_regression	13	DNAJC12	0.47572166834278445
Selected	Logistic Regression	model.logistic_regression	14	LINC00993	0.4749305923413496
Selected	Logistic Regression	model.logistic_regression	15	RGS22	0.4577415903892598
Selected	Logistic Regression	model.logistic_regression	16	ABAT	0.4566415816725219
Selected	Logistic Regression	model.logistic_regression	17	AGR3	0.4412802487004795
Selected	Logistic Regression	model.logistic_regression	18	AGR2	0.4381217948322408
Selected	Logistic Regression	model.logistic_regression	19	DEGS2	0.4211153364031078
Selected	Logistic Regression	model.logistic_regression	20	CT62	0.39810191561770725
Selected	Logistic Regression	model.logistic_regression	21	GATA3-AS1	0.3932681578635195
Selected	Logistic Regression	model.logistic_regression	22	ERBB4	0.3644766132195807
Selected	Logistic Regression	model.logistic_regression	23	SLC7A2	0.36303479437690345
Selected	Logistic Regression	model.logistic_regression	24	ADAMTS15	0.32751223252978884
Selected	Logistic Regression	model.logistic_regression	25	GREB1	0.31153260511021236
Selected	Logistic Regression	model.logistic_regression	26	KCNJ3	0.3029507469359344
Selected	Logistic Regression	model.logistic_regression	27	SLC16A6	0.2946876360060453
Selected	Logistic Regression	model.logistic_regression	28	TFF3	0.2917735655764524
Selected	Logistic Regression	model.logistic_regression	29	TNRC18P1	0.28076704921100326
Selected	Logistic Regression	model.logistic_regression	30	SLC39A6	0.25017335094441967
Comparison	Random Forest	model.random_forest	1	ESR1	0.1717206317423802
Comparison	Random Forest	model.random_forest	2	CCDC170	0.11684123188281723
Comparison	Random Forest	model.random_forest	3	C5AR2	0.10666979114006146
Comparison	Random Forest	model.random_forest	4	AGR3	0.0975229712657521
Comparison	Random Forest	model.random_forest	5	GATA3	0.0706523788576333
Comparison	Random Forest	model.random_forest	6	TBC1D9	0.04535318632341801
Comparison	Random Forest	model.random_forest	7	CT62	0.0419783760396283
Comparison	Random Forest	model.random_forest	8	THSD4	0.04050189933252878
Comparison	Random Forest	model.random_forest	9	DNAL1	0.03533571119816196
Comparison	Random Forest	model.random_forest	10	CYP2B7P	0.030789894344485716
Comparison	Random Forest	model.random_forest	11	ZMYND10	0.02811887289618082
Comparison	Random Forest	model.random_forest	12	POTEKP	0.020230101033764117
Comparison	Random Forest	model.random_forest	13	CA12	0.01926037264564226
Comparison	Random Forest	model.random_forest	14	MLPH	0.01884013968062428
Comparison	Random Forest	model.random_forest	15	LINC01087	0.01791630038559043
Comparison	Random Forest	model.random_forest	16	SLC39A6	0.015660189249979736
Comparison	Random Forest	model.random_forest	17	ABAT	0.01354543860411385
Comparison	Random Forest	model.random_forest	18	RGS22	0.010988526150581933
Comparison	Random Forest	model.random_forest	19	PGR	0.01084739881611983
Comparison	Random Forest	model.random_forest	20	NAT1	0.010488137386294647
Comparison	Random Forest	model.random_forest	21	DACH1	0.009675399870630895
Comparison	Random Forest	model.random_forest	22	PSAT1	0.008765583691982127
Comparison	Random Forest	model.random_forest	23	MAPT	0.008609752318555146
Comparison	Random Forest	model.random_forest	24	AGR2	0.008424371743472665
Comparison	Random Forest	model.random_forest	25	SLC7A2	0.008249410830522642
Comparison	Random Forest	model.random_forest	26	DEGS2	0.007100632455052536
Comparison	Random Forest	model.random_forest	27	GATA3-AS1	0.007076822583898385
Comparison	Random Forest	model.random_forest	28	SERPINA11	0.006532833761414625
Comparison	Random Forest	model.random_forest	29	TFF1	0.006340107008609886
Comparison	Random Forest	model.random_forest	30	SLC16A6	0.005963536760102242

Complete fold-level model feature stability

Model	Algorithm Id	Gene	Selection Frequency	Selected Folds	Total Folds
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Logistic Regression	model.logistic_regression	ESR1	1	9	9
Logistic Regression	model.logistic_regression	TBC1D9	1	9	9
Logistic Regression	model.logistic_regression	PGR	1	9	9
Logistic Regression	model.logistic_regression	CYP2B7P	1	9	9
Logistic Regression	model.logistic_regression	DACH1	1	9	9
Logistic Regression	model.logistic_regression	AGR3	0.8888888888888888	8	9
Logistic Regression	model.logistic_regression	DNALI1	0.8888888888888888	8	9
Logistic Regression	model.logistic_regression	POTEKP	0.8888888888888888	8	9
Logistic Regression	model.logistic_regression	LINC00993	0.8888888888888888	8	9
Logistic Regression	model.logistic_regression	CA12	0.8888888888888888	8	9
Logistic Regression	model.logistic_regression	TFF1	0.7777777777777778	7	9
Logistic Regression	model.logistic_regression	TNRC18P1	0.7777777777777778	7	9
Logistic Regression	model.logistic_regression	AGR2	0.7777777777777778	7	9
Logistic Regression	model.logistic_regression	SLC7A2	0.7777777777777778	7	9
Logistic Regression	model.logistic_regression	GATA3	0.7777777777777778	7	9
Logistic Regression	model.logistic_regression	KCNJ3	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	RGS22	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	ABAT	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	DNAJC12	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	DEGS2	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	SLC16A6	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	KCNK15	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	AFF3	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	GATA3-AS1	0.6666666666666666	6	9
Logistic Regression	model.logistic_regression	NAT1	0.5555555555555556	5	9
Logistic Regression	model.logistic_regression	ZMYND10	0.5555555555555556	5	9
Logistic Regression	model.logistic_regression	FSIP1	0.4444444444444444	4	9
Logistic Regression	model.logistic_regression	ADAMTS15	0.4444444444444444	4	9
Logistic Regression	model.logistic_regression	GREB1	0.4444444444444444	4	9
Logistic Regression	model.logistic_regression	SERPINA11	0.4444444444444444	4	9
Logistic Regression	model.logistic_regression	CLSTN2	0.4444444444444444	4	9
Logistic Regression	model.logistic_regression	SLC44A4	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	C5AR2	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	BMPR1B	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	FOXA1	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	STC2	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	CHAD	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	SLC39A6	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	THSD4	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	ERBB4	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	CHRD	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	CT62	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	MLPH	0.3333333333333333	3	9
Logistic Regression	model.logistic_regression	MAPT	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	GFRA1	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	TPRG1	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	CCDC170	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	CST9	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	AKR7A3	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	GP2	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	CPB1	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	KCNE4	0.2222222222222222	2	9
Logistic Regression	model.logistic_regression	SUSD3	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	SCUBE2	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	LINC01087	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	ANKRD30A	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	WNK4	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	PSAT1	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	STMND1	0.1111111111111111	1	9
Logistic Regression	model.logistic_regression	IL20	0.1111111111111111	1	9
Random Forest	model.random_forest	ESR1	1	9	9
Random Forest	model.random_forest	CCDC170	1	9	9
Random Forest	model.random_forest	AGR3	1	9	9
Random Forest	model.random_forest	GATA3	1	9	9
Random Forest	model.random_forest	C5AR2	1	9	9
Random Forest	model.random_forest	POTEKP	1	9	9
Random Forest	model.random_forest	FSIP1	1	9	9
Random Forest	model.random_forest	SLC39A6	1	9	9
Random Forest	model.random_forest	TBC1D9	1	9	9
Random Forest	model.random_forest	THSD4	1	9	9
Random Forest	model.random_forest	CT62	0.8888888888888888	8	9
Random Forest	model.random_forest	ZMYND10	0.8888888888888888	8	9
Random Forest	model.random_forest	LINC01087	0.8888888888888888	8	9
Random Forest	model.random_forest	GATA3-AS1	0.8888888888888888	8	9
Random Forest	model.random_forest	CA12	0.8888888888888888	8	9
Random Forest	model.random_forest	PSAT1	0.8888888888888888	8	9
Random Forest	model.random_forest	DNALI1	0.8888888888888888	8	9
Random Forest	model.random_forest	CYP2B7P	0.7777777777777778	7	9
Random Forest	model.random_forest	ABAT	0.7777777777777778	7	9
Random Forest	model.random_forest	NAT1	0.7777777777777778	7	9
Random Forest	model.random_forest	DEGS2	0.6666666666666666	6	9
Random Forest	model.random_forest	DACH1	0.6666666666666666	6	9
Random Forest	model.random_forest	RGS22	0.6666666666666666	6	9
Random Forest	model.random_forest	KCNK15	0.6666666666666666	6	9
Random Forest	model.random_forest	MLPH	0.5555555555555556	5	9
Random Forest	model.random_forest	PGR	0.5555555555555556	5	9
Random Forest	model.random_forest	AGR2	0.5555555555555556	5	9

Random Forest	model.random_forest	TFF1	0.5555555555555556	5	9
Random Forest	model.random_forest	SERPINA11	0.5555555555555556	5	9
Random Forest	model.random_forest	GREB1	0.4444444444444444	4	9
Random Forest	model.random_forest	AFF3	0.4444444444444444	4	9
Random Forest	model.random_forest	SCUBE2	0.4444444444444444	4	9
Random Forest	model.random_forest	MAPT	0.4444444444444444	4	9
Random Forest	model.random_forest	SLC44A4	0.3333333333333333	3	9
Random Forest	model.random_forest	ERBB4	0.3333333333333333	3	9
Random Forest	model.random_forest	SLC7A2	0.3333333333333333	3	9
Random Forest	model.random_forest	GFRA1	0.3333333333333333	3	9
Random Forest	model.random_forest	GRPR	0.3333333333333333	3	9
Random Forest	model.random_forest	CLSTN2	0.3333333333333333	3	9
Random Forest	model.random_forest	KCNJ3	0.2222222222222222	2	9
Random Forest	model.random_forest	STC2	0.2222222222222222	2	9
Random Forest	model.random_forest	SLC16A6	0.2222222222222222	2	9
Random Forest	model.random_forest	FOXA1	0.2222222222222222	2	9
Random Forest	model.random_forest	ADAMTS15	0.2222222222222222	2	9
Random Forest	model.random_forest	BMPR1B	0.2222222222222222	2	9
Random Forest	model.random_forest	GP2	0.1111111111111111	1	9
Random Forest	model.random_forest	CHRD	0.1111111111111111	1	9
Random Forest	model.random_forest	TNRC18P1	0.1111111111111111	1	9
Random Forest	model.random_forest	PRR15	0.1111111111111111	1	9
Random Forest	model.random_forest	EVL	0.1111111111111111	1	9
Random Forest	model.random_forest	SUSD3	0.1111111111111111	1	9
Random Forest	model.random_forest	TPRG1	0.1111111111111111	1	9
Random Forest	model.random_forest	CHAD	0.1111111111111111	1	9

Genes shared across all models

Gene	Scope
ABAT	shared across all models
AGR2	shared across all models
AGR3	shared across all models
C5AR2	shared across all models
CA12	shared across all models
CT62	shared across all models
CYP2B7P	shared across all models
DACH1	shared across all models
DEGS2	shared across all models
DNAL1	shared across all models
ESR1	shared across all models
GATA3	shared across all models
GATA3-AS1	shared across all models
PGR	shared across all models
POTEKP	shared across all models
RGS22	shared across all models
SLC16A6	shared across all models
SLC39A6	shared across all models
SLC7A2	shared across all models
TBC1D9	shared across all models

Cross-model model-specific genes

Model	Algorithm Id	Rank	Gene
Logistic Regression	model.logistic_regression	1	ADAMTS15
Logistic Regression	model.logistic_regression	2	DNAJC12
Logistic Regression	model.logistic_regression	3	ERBB4
Logistic Regression	model.logistic_regression	4	FOXA1
Logistic Regression	model.logistic_regression	5	GREB1
Logistic Regression	model.logistic_regression	6	KCNJ3
Logistic Regression	model.logistic_regression	7	LINC00993
Logistic Regression	model.logistic_regression	8	SLC44A4
Logistic Regression	model.logistic_regression	9	TFF3
Logistic Regression	model.logistic_regression	10	TNRC18P1
Random Forest	model.random_forest	1	CCDC170
Random Forest	model.random_forest	2	LINC01087
Random Forest	model.random_forest	3	MAPT
Random Forest	model.random_forest	4	MLPH
Random Forest	model.random_forest	5	NAT1
Random Forest	model.random_forest	6	PSAT1
Random Forest	model.random_forest	7	SERPINA11
Random Forest	model.random_forest	8	TFF1
Random Forest	model.random_forest	9	THSD4
Random Forest	model.random_forest	10	ZMYND10

Pairwise gene concordance

Pair	Model A Algorithm Id	Model A	Model B Algorithm Id	Model B	Shared Gene Count	Union Gene Count	Jaccard	Shared Genes
model.logistic_regression_vs_model.random_forest	model.logistic_regression	Logistic Regression	model.random_forest	Random Forest	20	40	0.5	["ABAT","AGR2","AGR3","C5AR2","CA12","CT62","CYP2B7P","DACH1","DEGS2","DNALI1","ESR1","GATA3","GATA3-AS1","PGR","POTEKP","RGS22","SLC16A6","SLC39A6","SLC7A2","TBC1D9"]

Pairwise ranking concordance

Pair	Model A Algorithm Id	Model A	Model B Algorithm Id	Model B	Common Ranked Gene Count	Spearman Rho Common Genes	Top N	Top N Overlap Count	Top N Jaccard
model.logistic_regression_vs_model.random_forest	model.logistic_regression	Logistic Regression	model.random_forest	Random Forest	20	0.48872180451127817	20	13	0.48148148148148145

Pairwise pathway concordance

Pair	Model A Algorithm Id	Model A	Model B Algorithm Id	Model B	Significant Pathway Jaccard	Shared Significant Pathways	Shared Significant Pathway Count
model.logistic_regression_vs_model.random_forest	model.logistic_regression	Logistic Regression	model.random_forest	Random Forest	0.8888888888888888	["Degradation of GABA","ESR-mediated signaling","Estrogen-dependent gene expression","Nuclear Receptor transcription pathway","Nuclear signaling by ERBB4","SUMOylation of intracellular receptors","Signaling by ERBB4","Signaling by Nuclear Receptors"]	8

Pathway enrichment by model

Model	Algorithm Id	Rank	Term	Fdr	Overlap Count	Overlap Genes
Logistic Regression	model.logistic_regression	1	Estrogen-dependent gene expression	3.195229661916671e-08	7	["ERBB4","ESR1","FOXA1","GATA3","GREB1","PGR","TFF3"]
Logistic Regression	model.logistic_regression	2	ESR-mediated signaling	4.08123872047086e-07	7	["ERBB4","ESR1","FOXA1","GATA3","GREB1","PGR","TFF3"]
Logistic Regression	model.logistic_regression	3	Signaling by Nuclear Receptors	2.8677097445771253e-06	7	["ERBB4","ESR1","FOXA1","GATA3","GREB1","PGR","TFF3"]
Logistic Regression	model.logistic_regression	4	Nuclear signaling by ERBB4	0.0004216896578549688	3	["ERBB4","ESR1","PGR"]
Logistic Regression	model.logistic_regression	5	Signaling by ERBB4	0.0020480328921905133	3	["ERBB4","ESR1","PGR"]
Logistic Regression	model.logistic_regression	6	SUMOylation of intracellular receptors	0.018515900845182256	2	["ESR1","PGR"]
Logistic Regression	model.logistic_regression	7	Degradation of GABA	0.04613074043199594	1	["ABAT"]
Logistic Regression	model.logistic_regression	8	Nuclear Receptor transcription pathway	0.04613074043199594	2	["ESR1","PGR"]
Logistic Regression	model.logistic_regression	9	SLC-mediated transmembrane transport	0.07404828103882972	3	["SLC39A6","SLC44A4","SLC7A2"]

Logistic Regression	model.logistic_regression	10	Constitutive Signaling by Aberrant PI3K in Cancer	0.07885733978008565	2	["ERBB4","ESR1"]
Logistic Regression	model.logistic_regression	11	RUNX1 regulates estrogen receptor mediated transcription	0.07885733978008565	1	["ESR1"]
Logistic Regression	model.logistic_regression	12	RUNX1 regulates transcription of genes involved in WNT signaling	0.07885733978008565	1	["ESR1"]
Logistic Regression	model.logistic_regression	13	Signal Transduction	0.07885733978008565	9	["C5AR2","ERBB4","ESR1","FOXA1","GATA3","GREB1","PGR","RGS2","TFF3"]
Logistic Regression	model.logistic_regression	14	Transmission across Chemical Synapses	0.07885733978008565	3	["ABAT","ERBB4","KCNJ3"]
Logistic Regression	model.logistic_regression	15	SLC-mediated bile acid transport	0.08580605801801884	1	["SLC44A4"]
Logistic Regression	model.logistic_regression	16	Downregulation of ERBB4 signaling	0.08737134002295395	1	["ERBB4"]
Logistic Regression	model.logistic_regression	17	Negative regulation of the PI3K/AKT network	0.08737134002295395	2	["ERBB4","ESR1"]
Logistic Regression	model.logistic_regression	18	PI3K events in ERBB4 signaling	0.08737134002295395	1	["ERBB4"]
Logistic Regression	model.logistic_regression	19	PI3K/AKT Signaling in Cancer	0.08737134002295395	2	["ERBB4","ESR1"]
Logistic Regression	model.logistic_regression	20	PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling	0.08737134002295395	2	["ERBB4","ESR1"]
Logistic Regression	model.logistic_regression	21	Zinc influx into cells by the SLC39 gene family	0.08737134002295395	1	["SLC39A6"]
Logistic Regression	model.logistic_regression	22	ERBB2 Activates PTK6 Signaling	0.09429796757415856	1	["ERBB4"]
Logistic Regression	model.logistic_regression	23	ERBB2 Regulates Cell Motility	0.09429796757415856	1	["ERBB4"]
Logistic Regression	model.logistic_regression	24	Formation of axial mesoderm	0.09429796757415856	1	["FOXA1"]
Logistic Regression	model.logistic_regression	25	GRB2 events in ERBB2 signaling	0.09429796757415856	1	["ERBB4"]
Logistic Regression	model.logistic_regression	26	Neuronal System	0.09429796757415856	3	["ABAT","ERBB4","KCNJ3"]
Logistic Regression	model.logistic_regression	27	PI3K events in ERBB2 signaling	0.09429796757415856	1	["ERBB4"]
Logistic Regression	model.logistic_regression	28	Reversible hydration of carbon dioxide	0.09429796757415856	1	["CA12"]
Logistic Regression	model.logistic_regression	29	SHC1 events in ERBB4 signaling	0.09429796757415856	1	["ERBB4"]
Logistic Regression	model.logistic_regression	30	TFAP2 (AP-2) family regulates transcription of growth factors and their receptors	0.09429796757415856	1	["ESR1"]
Logistic Regression	model.logistic_regression	31	Zinc transporters	0.09429796757415856	1	["SLC39A6"]
Logistic Regression	model.logistic_regression	32	Formation of the nephric duct	0.09622186610850925	1	["GATA3"]
Logistic Regression	model.logistic_regression	33	GABA synthesis, release, reuptake and degradation	0.09622186610850925	1	["ABAT"]

Logistic Regression	model.logistic_regression	34	Mitochondrial unfolded protein response (UPRmt)	0.09622186610850925	1	["ESR1"]
Logistic Regression	model.logistic_regression	35	SUMO E3 ligases SUMOylate target proteins	0.09622186610850925	2	["ESR1","PGR"]
Logistic Regression	model.logistic_regression	36	SUMOylation	0.09622186610850925	2	["ESR1","PGR"]
Logistic Regression	model.logistic_regression	37	SHC1 events in ERBB2 signaling	0.10532728630235233	1	["ERBB4"]
Logistic Regression	model.logistic_regression	38	Signaling by ERBB2 TMD/JMD mutants	0.10532728630235233	1	["ERBB4"]
Logistic Regression	model.logistic_regression	39	Long-term potentiation	0.10721565351599607	1	["ERBB4"]
Logistic Regression	model.logistic_regression	40	Neurotransmitter receptors and postsynaptic signal transmission	0.10890822770255049	2	["ERBB4","KCNJ3"]
Logistic Regression	model.logistic_regression	41	Signaling by ERBB2 KD Mutants	0.10890822770255049	1	["ERBB4"]
Logistic Regression	model.logistic_regression	42	Transcriptional regulation by RUNX1	0.10890822770255049	2	["ESR1","GATA3"]
Logistic Regression	model.logistic_regression	43	Activation of G protein gated Potassium channels	0.10937343477138134	1	["KCNJ3"]
Logistic Regression	model.logistic_regression	44	Downregulation of ERBB2 signaling	0.10937343477138134	1	["ERBB4"]
Logistic Regression	model.logistic_regression	45	G protein gated Potassium channels	0.10937343477138134	1	["KCNJ3"]
Logistic Regression	model.logistic_regression	46	Inhibition of voltage gated Ca2+ channels via Gbeta/gamma subunits	0.10937343477138134	1	["KCNJ3"]
Logistic Regression	model.logistic_regression	47	Signaling by ERBB2 in Cancer	0.10937343477138134	1	["ERBB4"]
Logistic Regression	model.logistic_regression	48	Synthesis of PC	0.10937343477138134	1	["SLC44A4"]
Logistic Regression	model.logistic_regression	49	Signaling by Receptor Tyrosine Kinases	0.11035558279004216	3	["ERBB4","ESR1","PGR"]
Logistic Regression	model.logistic_regression	50	Amino acid transport across the plasma membrane	0.11561545261887943	1	["SLC7A2"]
Random Forest	model.random_forest	1	Estrogen-dependent gene expression	0.002291630251617274	4	["ESR1","GATA3","PGR","TFF1"]
Random Forest	model.random_forest	2	ESR-mediated signaling	0.006956167849726013	4	["ESR1","GATA3","PGR","TFF1"]
Random Forest	model.random_forest	3	Signaling by Nuclear Receptors	0.01709508511505931	4	["ESR1","GATA3","PGR","TFF1"]
Random Forest	model.random_forest	4	Nuclear signaling by ERBB4	0.01977133876849524	2	["ESR1","PGR"]
Random Forest	model.random_forest	5	SUMOylation of intracellular receptors	0.01977133876849524	2	["ESR1","PGR"]
Random Forest	model.random_forest	6	Acetylation	0.035821914031101916	1	["NAT1"]
Random Forest	model.random_forest	7	Degradation of GABA	0.035821914031101916	1	["ABAT"]
Random Forest	model.random_forest	8	Nuclear Receptor transcription pathway	0.035821914031101916	2	["ESR1","PGR"]
Random Forest	model.random_forest	9	Signaling by ERBB4	0.035821914031101916	2	["ESR1","PGR"]

Random Forest	model.random_forest	10	RUNX1 regulates estrogen receptor mediated transcription	0.0832459441420726	1	["ESR1"]
Random Forest	model.random_forest	11	RUNX1 regulates transcription of genes involved in WNT signaling	0.0832459441420726	1	["ESR1"]
Random Forest	model.random_forest	12	Caspase-mediated cleavage of cytoskeletal proteins	0.12160665267101459	1	["MAPT"]
Random Forest	model.random_forest	13	Reversible hydration of carbon dioxide	0.12160665267101459	1	["CA12"]
Random Forest	model.random_forest	14	Serine metabolism	0.12160665267101459	1	["PSAT1"]
Random Forest	model.random_forest	15	Zinc influx into cells by the SLC39 gene family	0.12160665267101459	1	["SLC39A6"]
Random Forest	model.random_forest	16	TFAP2 (AP-2) family regulates transcription of growth factors and their receptors	0.1234244948497771	1	["ESR1"]
Random Forest	model.random_forest	17	Formation of the nephric duct	0.12977323635524765	1	["GATA3"]
Random Forest	model.random_forest	18	Mitochondrial unfolded protein response (UPRmt)	0.12977323635524765	1	["ESR1"]
Random Forest	model.random_forest	19	SUMO E3 ligases SUMOylate target proteins	0.12977323635524765	2	["ESR1","PGR"]
Random Forest	model.random_forest	20	SUMOylation	0.12977323635524765	2	["ESR1","PGR"]
Random Forest	model.random_forest	21	Zinc transporters	0.12977323635524765	1	["SLC39A6"]
Random Forest	model.random_forest	22	GABA synthesis, release, reuptake and degradation	0.1306693643242061	1	["ABAT"]
Random Forest	model.random_forest	23	Transcriptional regulation by RUNX1	0.15276229829298654	2	["ESR1","GATA3"]
Random Forest	model.random_forest	24	Activation of AMPK downstream of NMDARs	0.16505688692874643	1	["MAPT"]
Random Forest	model.random_forest	25	Amino acid transport across the plasma membrane	0.16505688692874643	1	["SLC7A2"]
Random Forest	model.random_forest	26	Apoptotic cleavage of cellular proteins	0.16505688692874643	1	["MAPT"]
Random Forest	model.random_forest	27	Defective B3GALT1 causes PpS	0.16505688692874643	1	["THSD4"]
Random Forest	model.random_forest	28	Deubiquitination	0.16505688692874643	2	["ESR1","GATA3"]
Random Forest	model.random_forest	29	O-glycosylation of TSR domain-containing proteins	0.16505688692874643	1	["THSD4"]
Random Forest	model.random_forest	30	Ovarian tumor domain proteases	0.16505688692874643	1	["ESR1"]
Random Forest	model.random_forest	31	Paracetamol ADME	0.16505688692874643	1	["NAT1"]
Random Forest	model.random_forest	32	SLC-mediated transmembrane transport	0.16505688692874643	2	["SLC39A6","SLC7A2"]

Random Forest	model.random_forest	33	SLC-mediated transport of amino acids	0.16505688692874643	1	["SLC7A2"]
Random Forest	model.random_forest	34	Sphingolipid de novo biosynthesis	0.16505688692874643	1	["DEGS2"]
Random Forest	model.random_forest	35	Transcriptional regulation by the AP-2 (TFAP2) family of transcription factors	0.16505688692874643	1	["ESR1"]
Random Forest	model.random_forest	36	Transmission across Chemical Synapses	0.16505688692874643	2	["ABAT","MAPT"]
Random Forest	model.random_forest	37	Regulation of MITF-M-dependent genes involved in pigmentation	0.1691435134291993	1	["MLPH"]
Random Forest	model.random_forest	38	Metal ion SLC transporters	0.1723062854104636	1	["SLC39A6"]
Random Forest	model.random_forest	39	Kidney development	0.17528700466973857	1	["GATA3"]
Random Forest	model.random_forest	40	Apoptotic execution phase	0.18773728137338944	1	["MAPT"]
Random Forest	model.random_forest	41	Neurotransmitter release cycle	0.18773728137338944	1	["ABAT"]
Random Forest	model.random_forest	42	HSP90 chaperone cycle for steroid hormone receptors (SHR) in the presence of ligand	0.20022581604402095	1	["PGR"]
Random Forest	model.random_forest	43	Regulation of RUNX2 expression and activity	0.20216398754432402	1	["ESR1"]
Random Forest	model.random_forest	44	Diseases associated with O-glycosylation of proteins	0.23587637309341455	1	["THSD4"]
Random Forest	model.random_forest	45	Extra-nuclear estrogen signaling	0.2429863855891241	1	["ESR1"]
Random Forest	model.random_forest	46	Differentiation of naive CD4+ T cells to T helper 2 cells (Th2 cells)	0.24372135979095222	1	["GATA3"]
Random Forest	model.random_forest	47	Constitutive Signaling by Aberrant PI3K in Cancer	0.24407045365797544	1	["ESR1"]
Random Forest	model.random_forest	48	Neuronal System	0.24407045365797544	2	["ABAT","MAPT"]
Random Forest	model.random_forest	49	PKR-mediated signaling	0.24407045365797544	1	["MAPT"]
Random Forest	model.random_forest	50	Post NMDA receptor activation events	0.24407045365797544	1	["MAPT"]

OOF prediction concordance

Pair	Model A Algorithm Id	Model A	Model B Algorithm Id	Model B	Common Oof Sample Count	Spearman Rho Oof Score	Mean Absolute Score Difference	Hard Class Agreement At 0.5
model.logistic_regression_vs_model.random_forest	model.logistic_regression	Logistic Regression	model.random_forest	Random Forest	622	0.7449023262713742	0.07538091317014327	0.9710610932475884

Highest model-disagreement samples

Sample Id	True Label	Score Range	Score Sd	Model Scores
Sample_001	1	0.7190929989251593	0.35954649946257966	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.26090700107484066}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.98}]
Sample_002	1	0.6544190015251403	0.32720950076257016	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2055809984748597}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.86}]
Sample_003	1	0.6534231935947932	0.32671159679739653	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.05657680640520683}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.71}]
Sample_004	1	0.5848305860349159	0.29241529301745794	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.25183608063175084}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.8366666666666668}]
Sample_005	0	0.5523684607663055	0.27618423038315276	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04259255632734982}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.68}]
Sample_006	1	0.5274074436726501	0.26370372183632507	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04259255632734982}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.57}]
Sample_007	1	0.5234895337146854	0.2617447668573427	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.45651046628531455}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.98}]
Sample_008	1	0.5074089379467523	0.2537044689733761	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.42592439538658095}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.9333333333333332}]
Sample_009	0	0.48848518838482924	0.24424259419241462	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.004848144948504085}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.49333333333333335}]
Sample_010	1	0.47450361238858296	0.23725180619429148	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.4621630542780837}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.9366666666666666}]
Sample_011	1	0.47341957522291606	0.23670978761145803	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.4699137581104173}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.9433333333333334}]
Sample_012	0	0.47329849549518666	0.23664924774759333	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03336817117148003}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.5066666666666667}]
Sample_013	0	0.44852649077479223	0.22426324538739612	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0714735092252078}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.52}]
Sample_014	1	0.44503234172485767	0.22251617086242884	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09830099160847568}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.5433333333333333}]
Sample_015	0	0.4446810002696546	0.22234050013482728	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.015318999730345355}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.45999999999999996}]
Sample_016	0	0.4442722231261106	0.22213611156305532	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04572777687388938}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.49}]
Sample_017	0	0.430675428953335	0.2153377144766675	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.22265790437999833}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.6533333333333333}]
Sample_018	0	0.424446501794377	0.21222325089718846	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03888683153895641}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.4633333333333334}]
Sample_019	1	0.4130230501555836	0.2065115250777918	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2136436165110831}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.6266666666666667}]
Sample_020	1	0.4061711850010024	0.2030855925005012	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2804954816656641}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.6866666666666665}]

Cross-model biological program support

Term	Support Model Count	Support Fraction	Support Tier	Supporting Models	Supporting Algorithm Ids	Supporting Genes	Best Fdr	Fdr By Model
Estrogen-dependent gene expression	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ERBB4", "ESR1", "FOXA1", "GATA3", "GREB1", "PGR", "TFF1", "TFF3"]	3.195229661916671e-08	{"model.logistic_regression":3.195229661916671e-08, "model.random_forest":0.002291630251617274}

ESR-mediated signaling	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ERBB4", "ESR1", "FOXA1", "GATA3", "GREB1", "PGR", "TFF1", "TFF3"]	4.08123872047086e-07	{"model.logistic_regression": 4.08123872047086e-07, "model.random_forest": 0.006956167849726013}
Signaling by Nuclear Receptors	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ERBB4", "ESR1", "FOXA1", "GATA3", "GREB1", "PGR", "TFF1", "TFF3"]	2.8677097445771253e-06	{"model.logistic_regression": 2.8677097445771253e-06, "model.random_forest": 0.01709508511505931}
Nuclear signaling by ERBB4	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ERBB4", "ESR1", "PGR"]	0.0004216896578549688	{"model.logistic_regression": 0.0004216896578549688, "model.random_forest": 0.01977133876849524}
Signaling by ERBB4	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ERBB4", "ESR1", "PGR"]	0.0020480328921905133	{"model.logistic_regression": 0.0020480328921905133, "model.random_forest": 0.035821914031101916}
SUMOylation of intracellular receptors	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ESR1", "PGR"]	0.018515900845182256	{"model.logistic_regression": 0.018515900845182256, "model.random_forest": 0.01977133876849524}
Degradation of GABA	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ABAT"]	0.035821914031101916	{"model.logistic_regression": 0.04613074043199594, "model.random_forest": 0.035821914031101916}
Nuclear Receptor transcription pathway	2	1	shared	["Logistic Regression", "Random Forest"]	["model.logistic_regression", "model.random_forest"]	["ESR1", "PGR"]	0.035821914031101916	{"model.logistic_regression": 0.04613074043199594, "model.random_forest": 0.035821914031101916}
Acetylation	1	0.5	model_specific	["Random Forest"]	["model.random_forest"]	["NAT1"]	0.035821914031101916	{"model.random_forest": 0.035821914031101916}

Patient-level prediction agreement — summary

Total patients	High agreement	Moderate disagreement	Strong disagreement	Rows shown in report
622	549	55	18	50

The report body shows at most 50 informative patients, prioritizing model disagreement/uncertainty. The complete patient-level table remains available in the Publication Package as [tables/patient_agreement.csv](#).

Patient-level prediction agreement — informative cases

Sample Id	Observed Class	Mean Oof Score	Score Sd	Score Range	Hard Class Agreement	Consensus Class At 0.5	Agreement Category	Model Scores
Sample_001	1	0.6204535005374203	0.35954649946257966	0.7190929989251593	0.5	0	strong_disagreement	{["algorithm_id": "model.logistic_regression", "model": "Logistic Regression", "score": 0.26090700107484066], ["algorithm_id": "model.random_forest", "model": "Random Forest", "score": 0.98]}
Sample_002	1	0.5327904992374298	0.32720950076257016	0.6544190015251403	0.5	0	strong_disagreement	{["algorithm_id": "model.logistic_regression", "model": "Logistic Regression", "score": 0.2055809984748597], ["algorithm_id": "model.random_forest", "model": "Random Forest", "score": 0.86]}
Sample_003	1	0.3832884032026034	0.32671159679739653	0.6534231935947932	0.5	0	strong_disagreement	{["algorithm_id": "model.logistic_regression", "model": "Logistic Regression", "score": 0.05657680640520683], ["algorithm_id": "model.random_forest", "model": "Random Forest", "score": 0.71]}
Sample_004	1	0.5442513736492088	0.29241529301745794	0.5848305860349159	0.5	0	strong_disagreement	{["algorithm_id": "model.logistic_regression", "model": "Logistic Regression", "score": 0.25183608063175084], ["algorithm_id": "model.random_forest", "model": "Random Forest", "score": 0.8366666666666666]}
Sample_005	0	0.4038157696168473	0.27618423038315276	0.5523684607663055	0.5	0	strong_disagreement	{["algorithm_id": "model.logistic_regression", "model": "Logistic Regression", "score": 0.1276315392336945], ["algorithm_id": "model.random_forest", "model": "Random Forest", "score": 0.68]}

Sample_006	1	0.306296278 1636749	0.263703721 83632507	0.527407443 6726501	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04259255632734982},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.57}}}
Sample_007	1	0.718255233 1426573	0.261744766 8573427	0.523489533 7146854	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.45651046628531455},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.98}}}
Sample_008	1	0.679628864 3599571	0.253704468 9733761	0.507408937 9467523	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.42592439538658095},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9333333333333332}}}
Sample_010	1	0.699414860 4723751	0.237251806 19429148	0.474503612 38858296	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.4621630542780837},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9366666666666666}}}
Sample_011	1	0.706623545 7218754	0.236709787 61145803	0.473419575 22291606	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.4699137581104173},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9433333333333334}}}
Sample_012	0	0.270017418 9190734	0.236649247 74759333	0.473298495 49518666	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03336817117148003},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.5066666666666667}}}
Sample_013	0	0.295736754 6126039	0.224263245 38739612	0.448526490 77479223	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0714735092252078},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.52}}}
Sample_014	1	0.320817162 4709045	0.222516170 86242884	0.445032341 72485767	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09830099160847568},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.5433333333333333}}}
Sample_017	0	0.437995618 8566658	0.215337714 4766675	0.430675428 953335	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.22265790437999833},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6533333333333333}}}
Sample_019	1	0.420155141 5888749	0.206511525 0777918	0.413023050 1555836	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2136436165110831},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6266666666666667}}}
Sample_020	1	0.483581074 16616536	0.203085592 5005012	0.406171185 0010024	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2804954816656641},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6866666666666665}}}
Sample_021	0	0.448707608 8504165	0.194625724 48291682	0.389251448 9658337	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2540818843674996},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6433333333333333}}}
Sample_022	1	0.315860705 0419148	0.190805961 62475189	0.381611923 24950377	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.12505474341716297},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.5066666666666667}}}
Sample_009	0	0.249090739 14091873	0.244242594 19241462	0.488485188 38482924	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.004848144948504085},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4933333333333333}}}
Sample_015	0	0.237659499 86517265	0.222340500 13482728	0.444681000 2696546	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.015318999730345355},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.45999999999999996}}}
Sample_016	0	0.267863888 4369447	0.222136111 56305532	0.444272223 1261106	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0457277687388938},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.49}}}
Sample_018	0	0.251110082 4361449	0.212223250 89718846	0.424446501 794377	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03888683153895641},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4633333333333334}}}
Sample_023	1	0.697797369 6236952	0.185535963 7096382	0.371071927 4192764	1	1	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.512261405914057},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.8833333333333334}}}
Sample_024	0	0.275679785 2621822	0.184320214 73781782	0.368640429 47563563	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09135957052436433},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.45999999999999996}}}
Sample_025	1	0.734512436 7993991	0.175487563 20060088	0.350975126 40120176	1	1	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5590248735987983},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.91}}}
Sample_026	1	0.794723821 8020661	0.175276178 19793395	0.350552356 3958679	1	1	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6194476436041322},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9700000000000001}}}
Sample_027	0	0.221879724 74973092	0.171453608 58360242	0.342907217 1672049	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.050426116166128486},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.3933333333333337}}}

Sample_028	1	0.745724671 276249	0.170941995 39041766	0.341883990 7808353	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5747826758858313},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9166666666666666}}]
Sample_029	1	0.180415110 6394214	0.166251556 02724528	0.332503112 05449057	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.014163554612176124},{ "algorithm_id" ":"model.random_forest","model":"Random Forest","score":0.3466666666666667}}]
Sample_030	0	0.168330867 79284764	0.165002465 54048567	0.330004931 08097134	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0033284022523619706},{ "algorithm_i d":"model.random_forest","model":"Random Forest","score":0.3333333333333333}}]
Sample_031	1	0.715710503 4109582	0.164289496 5890418	0.328578993 1780836	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5514210068219164},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.88}}]
Sample_032	0	0.174791646 1105314	0.158541687 22280187	0.317083374 4456038	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0162495888729534},{ "algorithm_id" ":"model.random_forest","model":"Random Forest","score":0.3333333333333333}}]
Sample_033	1	0.738531235 3745284	0.158135431 2921382	0.316270862 5842764	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5803958040823902},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.8966666666666666}}]
Sample_034	0	0.205215093 2844821	0.158118240 04885125	0.316236480 0977025	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04709685323563085},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.3633333333333333}}]
Sample_035	1	0.837118126 315902	0.156215207 01743125	0.312430414 0348625	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6809029192984708},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9933333333333333}}]
Sample_036	1	0.810856323 3579544	0.155810343 30871224	0.311620686 6174245	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6550459800492422},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9666666666666667}}]
Sample_037	0	0.252022469 9396711	0.154644196 72699564	0.309288393 4539913	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09737827321267543},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4066666666666667}}]
Sample_038	1	0.722485608 2921723	0.154181058 37449425	0.308362116 7489885	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5683045499176781},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.8766666666666666}}]
Sample_039	1	0.829158271 4592043	0.147508395 20746223	0.295016790 41492446	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6816498762517421},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9766666666666666}}]
Sample_040	1	0.795275607 327472	0.144724392 672528	0.289448785 345056	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.650551214654944},{ "algorithm_id":"m odel.random_forest","model":"Random Forest","score":0.9400000000000001}}]
Sample_041	0	0.175604888 20804424	0.144395111 7919558	0.288790223 58391154	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03120977641608844},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.32}}]
Sample_042	1	0.827271051 1277769	0.142728948 8722232	0.285457897 7444464	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6845421022555537},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9700000000000001}}]
Sample_043	1	0.798550975 6166767	0.141449024 38332335	0.282898048 7666467	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.657101951233534},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9400000000000001}}]
Sample_044	1	0.812149359 8503803	0.141183973 4829532	0.282367946 9659064	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6709653863674271},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9533333333333335}}]
Sample_045	0	0.269734743 82665164	0.136931922 8400151	0.273863845 6800302	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.13280282098663654},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4066666666666667}}]
Sample_046	0	0.150937199 06486128	0.135729467 6018054	0.271458935 2036108	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.01520773146305586},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.2866666666666667}}]
Sample_047	1	0.862481425 4695925	0.130851907 8637408	0.261703815 7274816	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.7316295176058517},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9933333333333333}}]
Sample_048	1	0.863654857 473345	0.129678475 85998826	0.259356951 7199765	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.7339763816133568},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9933333333333333}}]
Sample_049	1	0.820619722 5119896	0.129380277 48801023	0.258760554 97602046	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6912394450239794},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9499999999999999}}]

Sample_050	1	0.8308808863589121	0.12911911364108786	0.2582382272821757	1	1	moderate_disagreement	{["algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.7017617727178243},{["algorithm_id":"model.random_forest","model":"Random Forest","score":0.96}]}
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Patient expression pattern summaries

Agreement Category	Sample Count	Gene	Mean Z Expression
high agreement	549	LINC01087	0.09327604831759212
high agreement	549	POTEKP	0.09261498711004122
high agreement	549	RGS22	0.09006676370554759
high agreement	549	C5AR2	0.08492731523908399
high agreement	549	GATA3-AS1	0.08397212693633756
high agreement	549	ESR1	0.0817165517808697
high agreement	549	SERPINA11	0.07857579435047032
high agreement	549	THSD4	0.07853994924255422
high agreement	549	SLC39A6	0.07837997860370906
high agreement	549	NAT1	0.07644490427140702
high agreement	549	AGR3	0.0754816467102582
high agreement	549	TFF1	0.07332249047748664
moderate disagreement	55	POTEKP	-0.7137608142765172
moderate disagreement	55	LINC01087	-0.6866807334846133
moderate disagreement	55	AGR3	-0.672283206803109
moderate disagreement	55	RGS22	-0.6639311154275132
moderate disagreement	55	THSD4	-0.6618193686607774
moderate disagreement	55	SERPINA11	-0.6220515981412411
moderate disagreement	55	GATA3	-0.614228250049642
moderate disagreement	55	ESR1	-0.6122415583432339
moderate disagreement	55	C5AR2	-0.6025060844627865
moderate disagreement	55	NAT1	-0.5941192527155615
moderate disagreement	55	GATA3-AS1	-0.5931917568863563
moderate disagreement	55	CT62	-0.5806973970327938
strong disagreement	18	C5AR2	-0.7492923011557665
strong disagreement	18	GATA3-AS1	-0.7486195032944356
strong disagreement	18	LINC01087	-0.7467283435947049
strong disagreement	18	RGS22	-0.7183578847684619
strong disagreement	18	POTEKP	-0.6438212854557498
strong disagreement	18	SLC39A6	-0.6231696354817081
strong disagreement	18	ESR1	-0.6216167343789147
strong disagreement	18	CYP2B7P	-0.6075436152416267
strong disagreement	18	TFF1	-0.5963155516836847
strong disagreement	18	NAT1	-0.5162051969803305
strong disagreement	18	KCNJ3	-0.5038271169966635
strong disagreement	18	SERPINA11	-0.49584851114666684

Gene → pathway → patient links

Term	Support Model Count	Gene Count Used	Activity Vs Consensus Prediction Spearman	Activity Vs Model Disagreement Spearman
Estrogen-dependent gene expression	2	8	0.7299587394888022	-0.4065159558321066
ESR-mediated signaling	2	8	0.7299587394888022	-0.4065159558321066
Signaling by Nuclear Receptors	2	8	0.7299587394888022	-0.4065159558321066
Nuclear signaling by ERBB4	2	3	0.7012782173907308	-0.3728482583445614
Signaling by ERBB4	2	3	0.7012782173907308	-0.3728482583445614
SUMOylation of intracellular receptors	2	2	0.6946291821169494	-0.3799647446942445
Degradation of GABA	2	1	0.6517148395844691	-0.38188481469920693
Nuclear Receptor transcription pathway	2	2	0.6946291821169494	-0.3799647446942445
Acetylation	1	1	0.6881568746661886	-0.4132080759927643

Patient-level model disagreement

Sample Id	Observed Class	Mean Oof Score	Score Sd	Score Range	Hard Class Agreement	Consensus Class At 0.5	Agreement Category	Model Scores
Sample_001	1	0.6204535005374203	0.35954649946257966	0.7190929989251593	0.5	0	strong_disagreement	{["algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.26090700107484066},{["algorithm_id":"model.random_forest","model":"Random Forest","score":0.98}]}
Sample_002	1	0.5327904992374298	0.32720950076257016	0.6544190015251403	0.5	0	strong_disagreement	{["algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2055809984748597},{["algorithm_id":"model.random_forest","model":"Random Forest","score":0.86}]}
Sample_003	1	0.3832884032026034	0.32671159679739653	0.6534231935947932	0.5	0	strong_disagreement	{["algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.05657680640520683},{["algorithm_id":"model.random_forest","model":"Random Forest","score":0.71}]}

Sample_004	1	0.5442513736492088	0.29241529301745794	0.5848305860349159	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.25183608063175084},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.8366666666666666}}]
Sample_005	0	0.4038157696168473	0.27618423038315276	0.5523684607663055	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.1276315392336945},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.68}}]
Sample_006	1	0.3062962781636749	0.26370372183632507	0.5274074436726501	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04259255632734982},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.57}}]
Sample_007	1	0.7182552331426573	0.2617447668573427	0.5234895337146854	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.45651046628531455},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.98}}]
Sample_008	1	0.6796288643599571	0.2537044689733761	0.5074089379467523	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.42592439538658095},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9333333333333333}}]
Sample_009	0	0.24909073914091873	0.24424259419241462	0.48848518838482924	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.004848144948504085},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4933333333333333}}]
Sample_010	1	0.6994148604723751	0.23725180619429148	0.47450361238858296	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.4621630542780837},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9366666666666666}}]
Sample_011	1	0.7066235457218754	0.23670978761145803	0.47341957522291606	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.4699137581104173},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.9433333333333333}}]
Sample_012	0	0.2700174189190734	0.23664924774759333	0.47329849549518666	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03336817117148003},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.5066666666666667}}]
Sample_013	0	0.2957367546126039	0.22426324538739612	0.44852649077479223	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0714735092252078},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.52}}]
Sample_014	1	0.3208171624709045	0.22251617086242884	0.44503234172485767	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09830099160847568},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.5433333333333333}}]
Sample_015	0	0.23765949986517265	0.22234050013482728	0.4446810002696546	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.015318999730345355},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4599999999999999}}]
Sample_016	0	0.2678638884369447	0.22213611156305532	0.4442722231261106	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04572777687388938},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.49}}]
Sample_017	0	0.4379956188566658	0.2153377144766675	0.430675428953335	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.22265790437999833},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6533333333333333}}]
Sample_018	0	0.2511100824361449	0.21222325089718846	0.424446501794377	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03888683153895641},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4633333333333333}}]
Sample_019	1	0.4201551415888749	0.2065115250777918	0.4130230501555836	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2136436165110831},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6266666666666667}}]
Sample_020	1	0.48358107416616536	0.2030855925005012	0.4061711850010024	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2804954816656641},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6866666666666666}}]
Sample_021	0	0.4487076088504165	0.19462572448291682	0.3892514489658337	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.2540818843674996},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.6433333333333333}}]
Sample_022	1	0.3158607050419148	0.19080596162475189	0.38161192324950377	0.5	0	strong_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.12505474341716297},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.5066666666666667}}]
Sample_023	1	0.6977973696236952	0.1855359637096382	0.3710719274192764	1	1	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.512261405914057},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.8833333333333333}}]
Sample_024	0	0.2756797852621822	0.18432021473781782	0.36864042947563563	1	0	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09135957052436433},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.4599999999999999}}]
Sample_025	1	0.7345124367993991	0.17548756320060088	0.35097512640120176	1	1	moderate_disagreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5590248735987983},{ "algorithm_id": "model.random_forest","model":"Random Forest","score":0.91}}]

Sample_026	1	0.794723821 8020661	0.175276178 19793395	0.350552356 3958679	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6194476436041322},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9700000000000001}}
Sample_027	0	0.221879724 74973092	0.171453608 58360242	0.342907217 1672049	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.050426116166128486},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.3933333333333337}}
Sample_028	1	0.745724671 276249	0.170941995 39041766	0.341883990 7808353	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5747826758858313},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9166666666666666}}
Sample_029	1	0.180415110 6394214	0.166251556 02724528	0.332503112 05449057	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.014163554612176124},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.3466666666666667}}
Sample_030	0	0.168330867 79284764	0.165002465 54048567	0.330004931 08097134	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0033284022523619706},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.3333333333333333}}
Sample_031	1	0.715710503 4109582	0.164289496 5890418	0.328578993 1780836	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5514210068219164},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.88}}
Sample_032	0	0.174791646 1105314	0.158541687 22280187	0.317083374 4456038	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.016249958887729534},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.3333333333333333}}
Sample_033	1	0.738531235 3745284	0.158135431 2921382	0.316270862 5842764	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5803958040823902},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.8966666666666666}}
Sample_034	0	0.205215093 2844821	0.158118240 04885125	0.316236480 0977025	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.04709685323563085},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.3633333333333334}}
Sample_035	1	0.837118126 315902	0.156215207 01743125	0.312430414 0348625	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6809029192984708},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9933333333333333}}
Sample_036	1	0.810856323 3579544	0.155810343 30871224	0.311620686 6174245	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6550459800492422},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9666666666666667}}
Sample_037	0	0.252022469 9396711	0.154644196 72699564	0.309288393 4539913	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.09737827321267543},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.40666666666666673}}
Sample_038	1	0.722485608 2921723	0.154181058 37449425	0.308362116 7489885	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.5683045499176781},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.8766666666666666}}
Sample_039	1	0.829158271 4592043	0.147508395 20746223	0.295016790 41492446	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6816498762517421},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9766666666666666}}
Sample_040	1	0.795275607 327472	0.144724392 672528	0.289448785 345056	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.650551214654944},{ "algorithm_id":"m odel.random_forest","model":"Random Forest","score":0.9400000000000001}}
Sample_041	0	0.175604888 20804424	0.144395111 7919558	0.288790223 58391154	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.03120977641608844},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.32}}
Sample_042	1	0.827271051 1277769	0.142728948 8722232	0.285457897 7444464	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.684542102255537},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9700000000000001}}
Sample_043	1	0.798550975 6166767	0.141449024 38332335	0.282898048 7666467	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.657101951233534},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9400000000000001}}
Sample_044	1	0.812149359 8503803	0.141183973 4829532	0.282367946 9659064	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6709653863674271},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9533333333333333}}
Sample_045	0	0.269734743 82665164	0.136931922 8400151	0.273863845 6800302	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.13280282098663654},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.40666666666666673}}
Sample_046	0	0.150937199 06486128	0.135729467 6018054	0.271458935 2036108	1	0	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.0152077314630558},{ "algorithm_id" :"model.random_forest","model":"Random Forest","score":0.2866666666666667}}
Sample_047	1	0.862481425 4695925	0.130851907 8637408	0.261703815 7274816	1	1	moderate_dis agreement	{{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.7316295176058517},{ "algorithm_id":" model.random_forest","model":"Random Forest","score":0.9933333333333333}}

Sample_048	1	0.863654857473345	0.12967847585998826	0.2593569517199765	1	1	moderate_disagreement	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.7339763816133568}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.9933333333333333}]
Sample_049	1	0.8206197225119896	0.12938027748801023	0.25876055497602046	1	1	moderate_disagreement	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.6912394450239794}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.9499999999999999}]
Sample_050	1	0.8308808863589121	0.12911911364108786	0.2582382272821757	1	1	moderate_disagreement	[{"algorithm_id":"model.logistic_regression","model":"Logistic Regression","score":0.7017617727178243}, {"algorithm_id":"model.random_forest","model":"Random Forest","score":0.96}]

These analyses are descriptive evidence of convergence or disagreement and do not establish a biological mechanism, subtype, or causal relationship.

Patient-level agreement uses repeated-CV out-of-fold predictions within the outer-training cohort; the untouched holdout is not used.

Patient expression prototypes are descriptive group summaries and must not be interpreted as newly discovered clinical subtypes without independent validation.

Pathway support depends on the configured local pathway resource and over-representation analysis of each model-selected gene set.

Figure 5. ROC curve

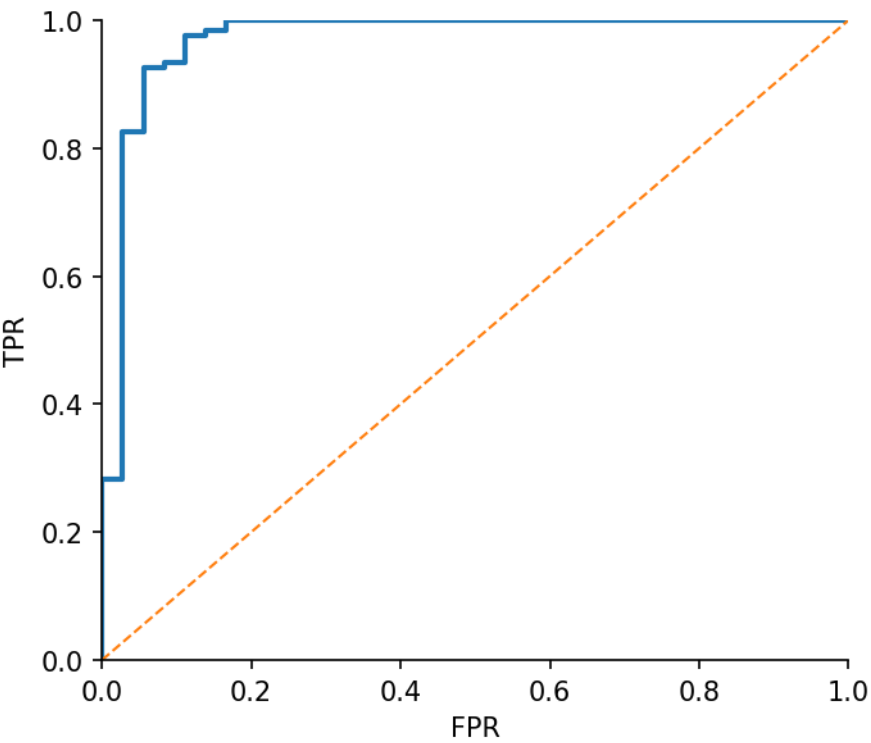


Figure 6. PR curve

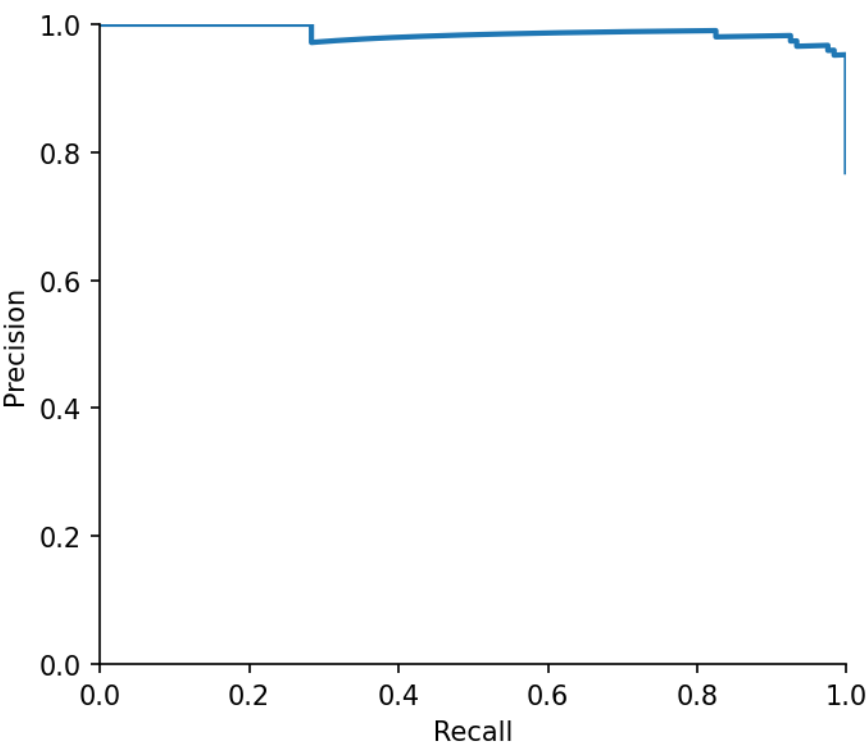


Table 3. Confusion matrix

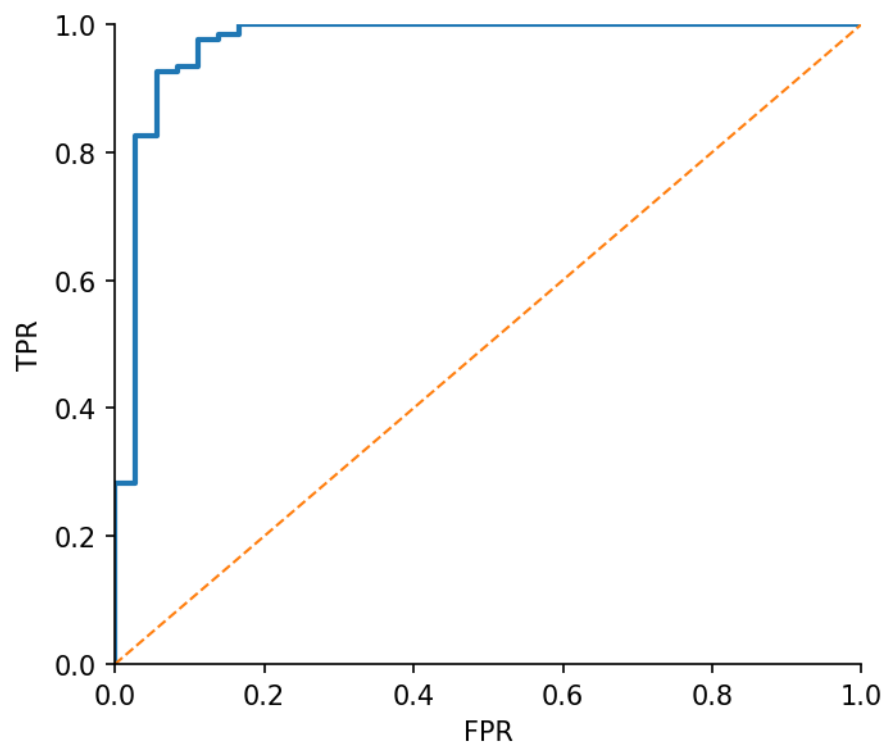
Actual class	Pred 0	Pred 1
Actual 0	32	4
Actual 1	4	116

5. Validation & Clinical Utility

Internal validation

ROC AUC	PR-AUC	Brier
0.970	0.987	0.041

Figure 7A. Internal validation ROC curve



Independent external validation

External N	Overlap features	External AUC	External PR-AUC	External accuracy	Sensitivity	Specificity
116	30	0.967	0.996	0.784	0.757	1.000

Threshold provenance & portability

Threshold role	Threshold	Status	Accuracy	Sensitivity	Specificity	Observed +	Predicted +	Confirmatory eligible
Reference / prespecified	0.500	severe_shift	0.784	0.757	1.000	0.888	0.672	Yes
Exploratory / user-adjusted	0.100	stable	0.931	0.922	1.000	0.888	0.819	No

The exploratory threshold result is a reportable research finding/sensitivity analysis, but it does not replace the prespecified reference threshold and a STABLE exploratory status does not upgrade confirmatory threshold-portability evidence.

Figure 7B. External validation ROC curve

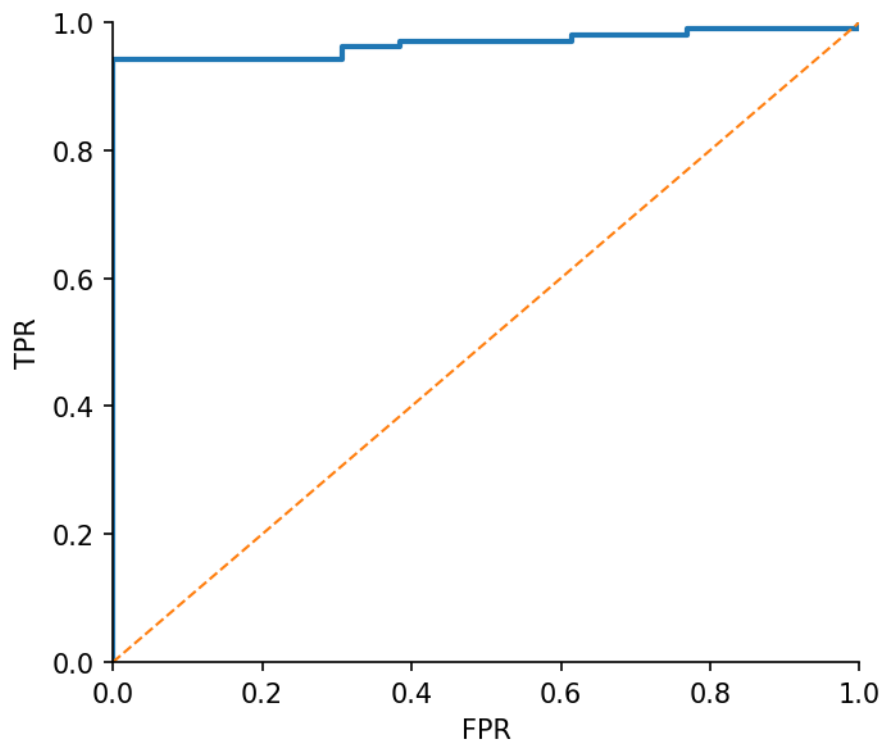
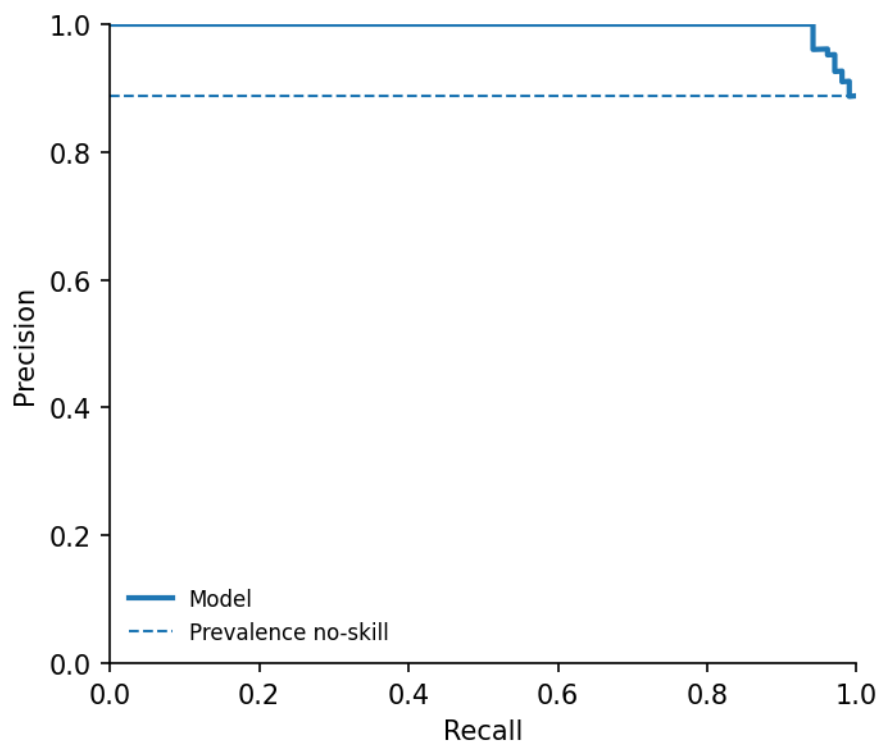


Figure 7C. External validation Precision–Recall curve



The dashed line is the PR no-skill baseline defined by external prevalence. PR-AUC should be interpreted relative to this baseline.

Figure 7D. External calibration curve

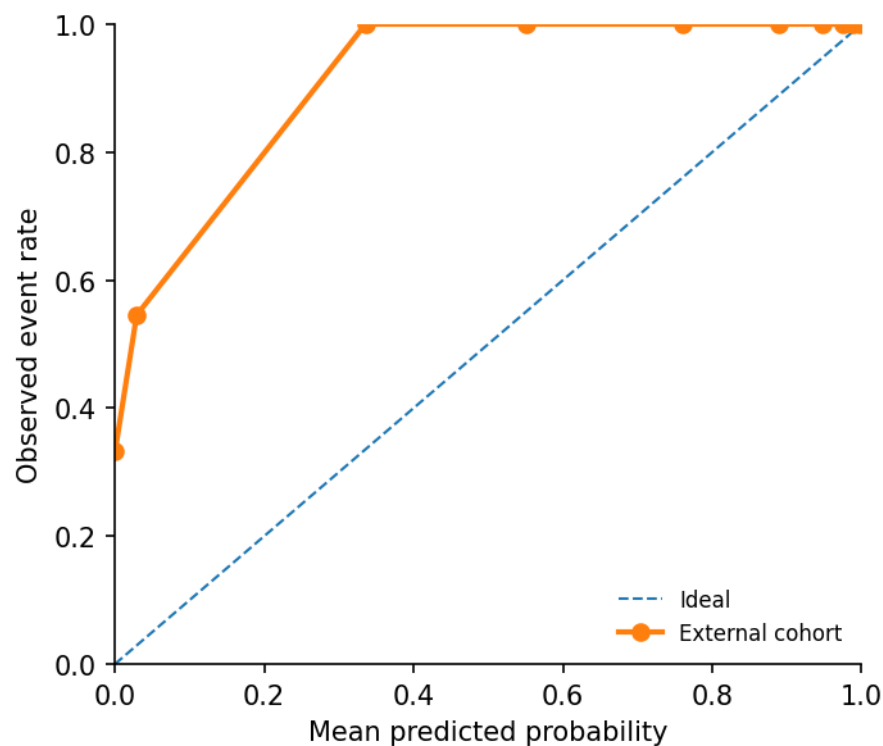


Table 4C-cal. External calibration diagnostics

Brier	Null Brier	Brier skill	Calibration intercept	Calibration slope	Curve bins
0.156	0.100	-0.564	4.305	0.665	10

Ideal calibration has intercept 0 and slope 1. Brier skill > 0 indicates improvement over a null probability model that predicts the external prevalence for every sample.

Table 4C-prob. Probability distribution & saturation

Min	Q1	Median	Q3	Max	≥0.90	≥0.99	≥0.999	≤0.10	Unique scores	Polarity audit
0.000	0.374	0.852	0.974	0.999	0.448	0.155	0.017	0.181	116	pass

Figure 7E. Decision curve analysis

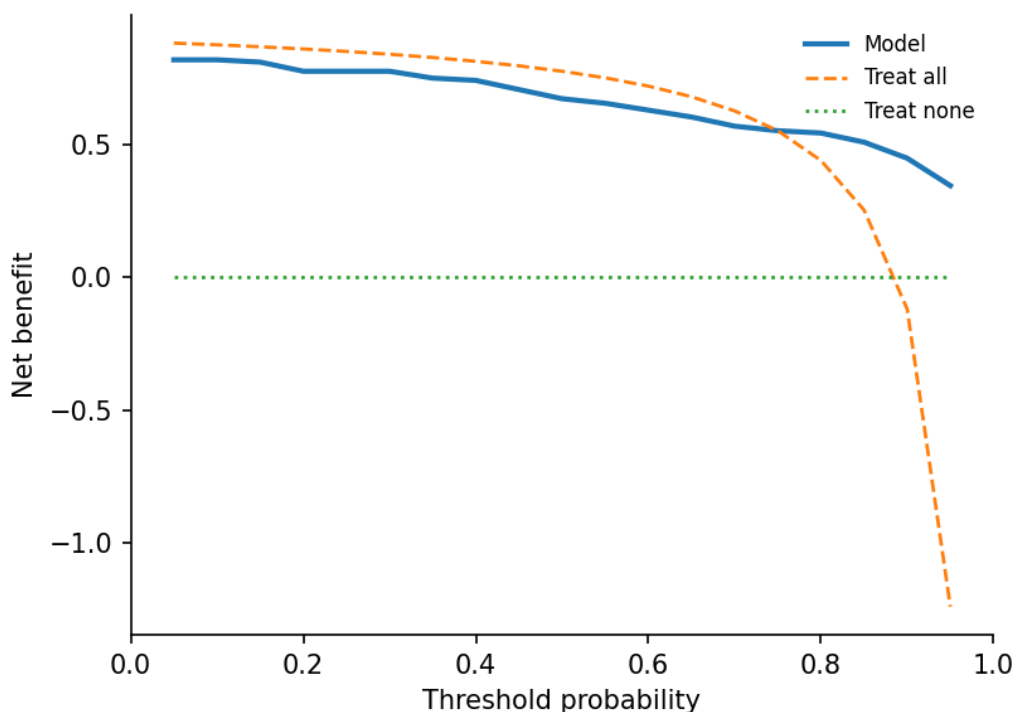


Table 4C-dca. Decision-analytic net benefit summary

Thresholds evaluated	Model best	> Treat none	> Treat all	Clinical-utility claim
19	4	19	4	No — decision-analytic only

Threshold role	Threshold	Model net benefit	Treat all	Treat none	Confirmatory threshold role
Reference / prespecified	0.50	0.672	0.776	0.000	Yes
Exploratory / user-adjusted	0.10	0.819	0.875	0.000	No

Decision Curve Analysis is decision-analytic evidence for the frozen external scores. It is not proof of clinical utility, and BioFlow does not use external outcomes to replace the prespecified threshold.

Table 4C. Reference / prespecified confusion matrix · threshold 0.50

Actual class	Pred 0	Pred 1
Actual 0	13	0
Actual 1	25	78

This confusion matrix corresponds to the formal reference/prespecified threshold used for confirmatory threshold-portability interpretation.

Table 4D. Exploratory / user-adjusted confusion matrix · threshold 0.10

Actual class	Pred 0	Pred 1
Actual 0	13	0
Actual 1	8	95

This result is an exploratory sensitivity analysis/research finding; even when STABLE, it does not replace confirmatory evidence at the reference threshold.

Table 4E. Threshold reclassification · exploratory analysis

N	Reclassified	Reclassified %	0 → 1	1 → 0	Unchanged	Corrected	Worsened	Net correct
116	17	0.147	17	0	99	17	0	17

This table describes sample reclassification from the reference/prespecified threshold to the exploratory/user-adjusted threshold. It is reportable exploratory evidence and does not validate or replace the prespecified threshold.

Transportability Matrix / Stability Score

The Stability Score is a descriptive transportability composite for frozen models on the same independent external cohort. It is not a clinical score and is never used to reselect the Modeling winner; exploratory/user-adjusted thresholds are excluded completely.

Winner score	Winner tier	Median score	Q1	Q3	Internal↔External rank p	Scored / total
—	INDETERMINATE	—	—	—	1.000	0/2

Role	Internal rank	Model	Internal AUC	External rank*	External AUC	AUC retention	Disc	Cal	Ref thr	Shift	Rep/feat	Protocol	Portability	Stability /100	Tier	Evidence grade
Winner	1	Logistic Regression	0.964	1	0.967	100.0 %	96.1	20.2	25.0	—	40.0	DIAGNOSTIC_ONLY	PARTIALLY_RETAINED	—	INDETERMINATE	DIAGNOSTIC_ONLY
Candidate	2	Random Forest	0.953	2	0.948	98.9 %	93.4	54.7	100.0	—	40.0	DIAGNOSTIC_ONLY	RETAINED	—	INDETERMINATE	DIAGNOSTIC_ONLY

Weights: Discrimination 35% · Calibration 20% · reference-threshold portability 20% · distribution shift 15% · representation/feature integrity 10%. Missing critical evidence yields INDETERMINATE; INVALID protocols do not receive an interpretable Stability Score.

Cross-model summary: Internal and external model ordering is similar on this cohort.

External Multi-Model Benchmark

The Modeling winner is locked to model.logistic_regression. External ranks are descriptive transportability observations for the current frozen models on the same independent cohort; they are not used to reselect, activate, or refit any model.

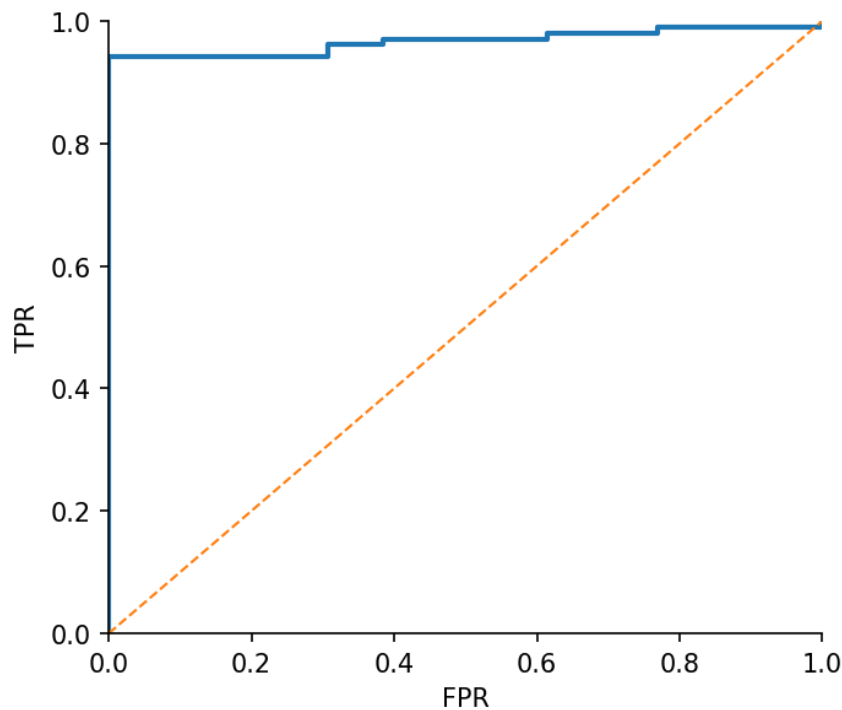
Role	Internal CV rank	Model	Internal CV AUC	External AUC rank*	External AUC	PR-AUC	Brier	Brier skill	Bal · acc	Sens	Spec	Ref thr	Thr status	Protocol	Portability	Features	CM TN/FP/FN/TP	Status
Winner	1	Logistic Regression	0.964	1	0.967	0.996	0.156	-0.564	0.879	0.757	1.000	0.50	severe_shift	DIAGNOSTIC_ONLY	PARTIALLY_RETAINED	30/30	13/0/25/78	OK
Candidate	2	Random Forest	0.953	2	0.948	0.994	0.070	0.301	0.807	0.922	0.692	0.50	stable	DIAGNOSTIC_ONLY	RETAINED	30/30	9/4/8/95	OK

* External AUC rank is descriptive only and does not change the Modeling winner selected by repeated-CV before external outcomes were examined.

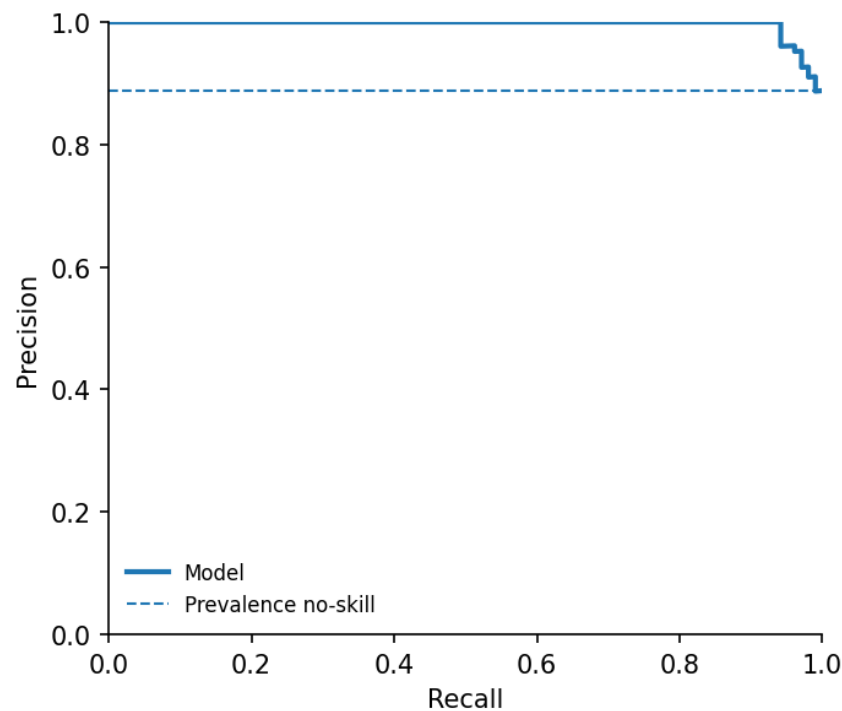
Logistic Regression · Modeling winner

External AUC	PR-AUC	Brier	Brier skill	Balanced accuracy	Sensitivity	Specificity	Reference threshold	Threshold status
0.967	0.996	0.156	-0.564	0.879	0.757	1.000	0.50	severe_shift

Logistic Regression · External ROC



Logistic Regression · External Precision–Recall



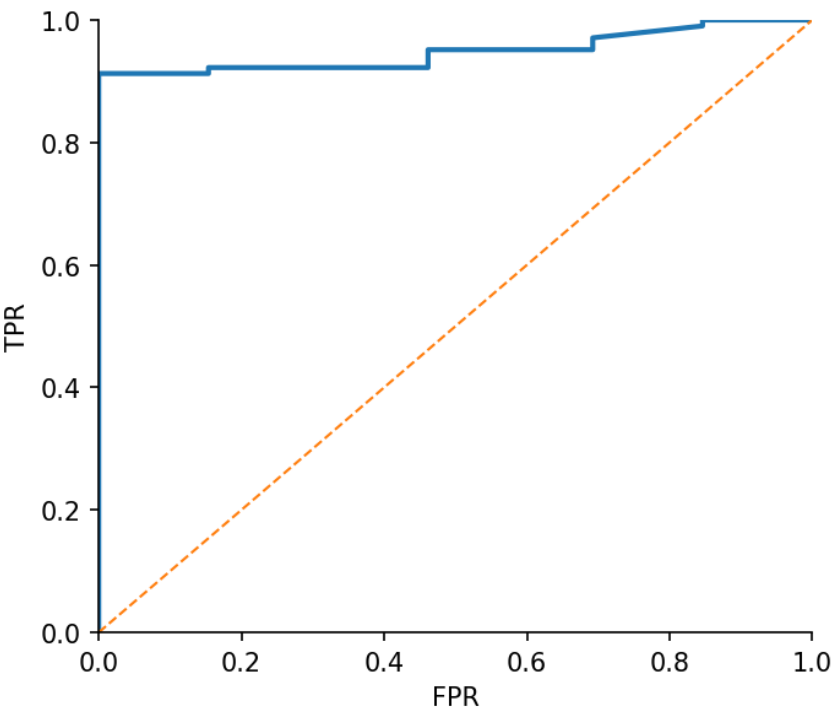
Logistic Regression · Reference-threshold confusion matrix

Actual class	Pred 0	Pred 1
Actual 0	13	0
Actual 1	25	78

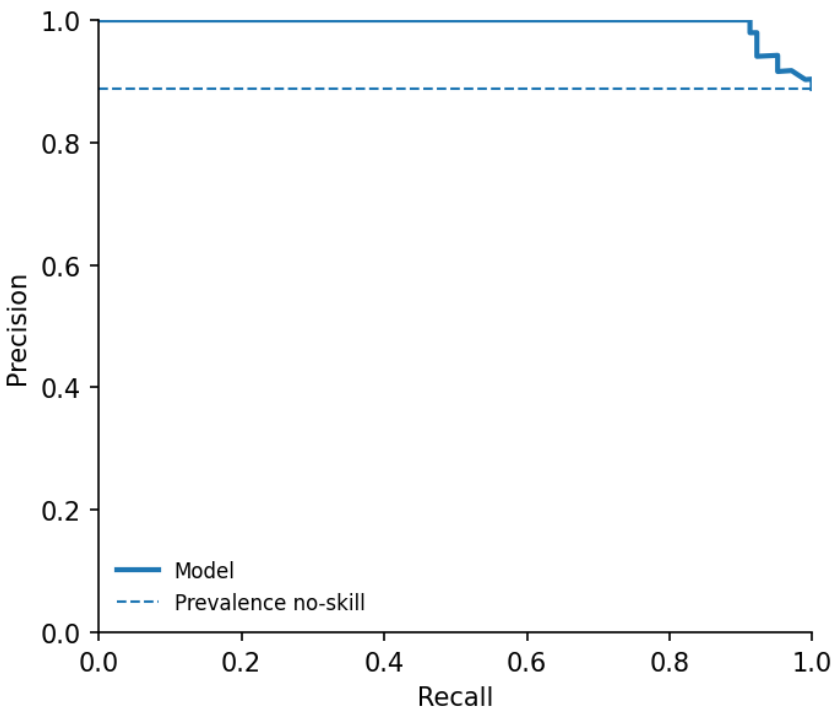
Random Forest · Frozen candidate

External AUC	PR-AUC	Brier	Brier skill	Balanced accuracy	Sensitivity	Specificity	Reference threshold	Threshold status
0.948	0.994	0.070	0.301	0.807	0.922	0.692	0.50	stable

Random Forest · External ROC



Random Forest · External Precision–Recall



Random Forest · Reference-threshold confusion matrix

Actual class	Pred 0	Pred 1
Actual 0	9	4
Actual 1	8	95

6. Biological Interpretation

Figure 8. Heatmap preview

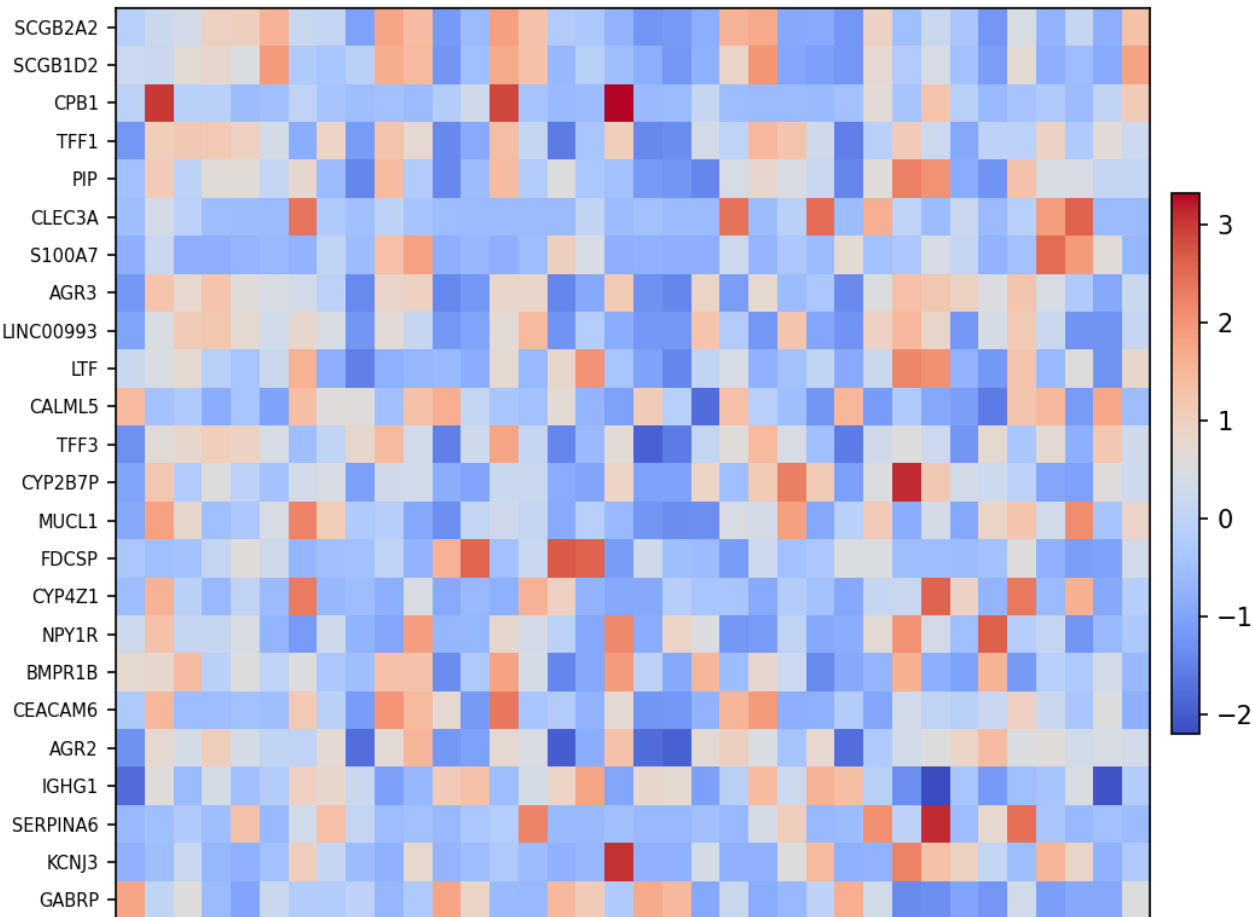


Table 5. Signature scores

Sample	Label	Score
Sample_046	0	1.850
Sample_051	0	1.791
Sample_052	0	1.758
Sample_018	0	1.752
Sample_034	0	1.743
Sample_053	0	1.740
Sample_054	0	1.724
Sample_055	0	1.721
Sample_015	0	1.718
Sample_056	0	1.713
Sample_057	0	1.711
Sample_058	1	1.706

7. AI Scientist Summary

Simulation Lab

Sample	Before	After	Delta
Sample_059	1.582	1.582	-0.000
Sample_060	1.291	1.291	-0.000
Sample_061	1.155	1.155	-0.000
Sample_062	1.330	1.330	-0.000
Sample_032	1.250	1.250	-0.000
Sample_063	1.589	1.589	-0.000
Sample_064	-1.018	-1.018	0.000
Sample_065	1.482	1.482	-0.000
Sample_066	1.301	1.301	-0.000
Sample_067	-1.083	-1.083	0.000
Sample_068	1.184	1.184	-0.000
Sample_054	1.724	1.724	-0.000

Evidence chain

Evidence	Strength	Finding
curated_gene_prior	high	ESR1 has a curated pathway entry in BioFlow knowledge base.
uploaded_data_match	moderate	ESR1 was found in the uploaded expression matrix as ESR1.
coexpression	moderate	Strongest positive expression neighbors: CCDC170 $r=0.87$, AGR3 $r=0.85$, THSD4 $r=0.85$, TBC1D9 $r=0.85$, C5AR2 $r=0.84$.
module_overlap	moderate	Disease-module overlap detected: er pr luminal program (FOXA1, GATA3, XBP1).
pathway_hypothesis	hypothesis	Candidate pathway mechanisms: ER/PR hormone-receptor signaling, luminal differentiation program, endocrine-therapy response biology.
phenotype_hypothesis	hypothesis	Candidate phenotype consequences: estrogen-dependent tumor growth, luminal breast cancer identity, endocrine sensitivity or resistance, estrogen-dependent growth.
related_genes	supporting	Related genes for validation: CCND1, FOXA1, GATA3, GREB1, PGR, TFF1, AR, XBP1.
curated_literature_prior	high	ESR1 is a key luminal breast cancer transcriptional regulator and endocrine-response context marker.

8. Methods

Expression matrices and metadata were parsed and quality-controlled before feature selection, candidate ranking, machine-learning modeling, and internal validation. When an independent external cohort was available, external validation was reported separately from internal performance. Heatmap visualization, signature scoring, perturbation simulation, and AI-assisted interpretation were integrated as complementary evidence layers.

9. Limitations

These results may be affected by sample size, cohort heterogeneity, batch effects, feature-selection strategy, and model overfitting. Even when one independent external cohort has been evaluated, broader transportability still depends on additional independent cohorts, platform diversity, and experimental or clinical evidence appropriate to the claim.

10. Integrated narrative

BioFlow Integrated Scientific Report

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1. Dataset Overview

Expression: TCGA_BRCA_ER_log2TPM1_expression.csv (22184 × 779)

Metadata: TCGA_BRCA_ER_metadata.csv (778 × 5)

2. QC Summary

Genes: 22184; Samples: 778; Missing fraction: 0.0000

Top variable genes: SCGB2A2, SCGB1D2, CPB1, TFF1, PIP, CLEC3A, S100A7, AGR3

3. Feature Selection · Methods and Candidate Features

Method: Welch delta score

Configuration: top_n=50; variance_threshold=0; random_seed=7; scope=training_subset_only

Candidate features: 50

Top-ranked features: ESR1, AGR3, TFF1, CYP2B7P, NAT1, SCUBE2, POTEKP, GFRA1, CCDC170, LINC01087, CA12, SERPINA11, FSIP1, C5AR2, DACH1, DNALI1, SLC39A6, THSD4, TFF3, SLC7A2, TBC1D9, CLSTN2, GATA3-AS1, SLC16A6, AGR2, AFF3, PGR, MAPT, ERBB4, GREB1, GATA3, CT62, SUSD3, TNRC18P1, LINC00993, TPRG1, ZMYND10, DNAJC12, ABAT, KCNJ3, PSAT1, DEGS2, GRPR, ADAMTS15, KCNE4, ANKRD30A, MLPH, RGS22, FOXA1, SLC44A4

Fold-level candidate-feature stability:

- ESR1: selection frequency=1.000 (9/9 folds)
- AGR3: selection frequency=1.000 (9/9 folds)
- TFF1: selection frequency=1.000 (9/9 folds)
- CYP2B7P: selection frequency=1.000 (9/9 folds)
- NAT1: selection frequency=1.000 (9/9 folds)
- SCUBE2: selection frequency=1.000 (9/9 folds)
- FSIP1: selection frequency=1.000 (9/9 folds)
- CCDC170: selection frequency=1.000 (9/9 folds)
- GFRA1: selection frequency=1.000 (9/9 folds)
- POTEKP: selection frequency=1.000 (9/9 folds)
- CA12: selection frequency=1.000 (9/9 folds)
- AFF3: selection frequency=1.000 (9/9 folds)
- SERPINA11: selection frequency=1.000 (9/9 folds)
- SLC39A6: selection frequency=1.000 (9/9 folds)

- C5AR2: selection frequency=1.000 (9/9 folds)
- TFF3: selection frequency=1.000 (9/9 folds)
- LINC01087: selection frequency=1.000 (9/9 folds)
- THSD4: selection frequency=1.000 (9/9 folds)
- TBC1D9: selection frequency=1.000 (9/9 folds)
- SLC7A2: selection frequency=1.000 (9/9 folds)
- DACH1: selection frequency=1.000 (9/9 folds)
- SLC16A6: selection frequency=1.000 (9/9 folds)
- DNALI1: selection frequency=1.000 (9/9 folds)
- GREB1: selection frequency=1.000 (9/9 folds)
- GATA3: selection frequency=1.000 (9/9 folds)
- GATA3-AS1: selection frequency=1.000 (9/9 folds)
- CLSTN2: selection frequency=1.000 (9/9 folds)
- MAPT: selection frequency=1.000 (9/9 folds)
- PGR: selection frequency=1.000 (9/9 folds)
- ABAT: selection frequency=1.000 (9/9 folds)
- ZMYND10: selection frequency=1.000 (9/9 folds)
- DEGS2: selection frequency=1.000 (9/9 folds)
- AGR2: selection frequency=1.000 (9/9 folds)
- ERBB4: selection frequency=1.000 (9/9 folds)
- LINC00993: selection frequency=1.000 (9/9 folds)
- KCNJ3: selection frequency=0.889 (8/9 folds)
- TPRG1: selection frequency=0.889 (8/9 folds)
- CT62: selection frequency=0.889 (8/9 folds)
- SUS3: selection frequency=0.889 (8/9 folds)
- PSAT1: selection frequency=0.889 (8/9 folds)
- TNRC18P1: selection frequency=0.889 (8/9 folds)
- DNAJC12: selection frequency=0.778 (7/9 folds)
- RGS22: selection frequency=0.667 (6/9 folds)
- KCNE4: selection frequency=0.667 (6/9 folds)
- ADAMTS15: selection frequency=0.667 (6/9 folds)
- KCNK15: selection frequency=0.667 (6/9 folds)
- MLPH: selection frequency=0.556 (5/9 folds)
- AKR7A3: selection frequency=0.556 (5/9 folds)
- GRPR: selection frequency=0.556 (5/9 folds)
- ANKRD30A: selection frequency=0.556 (5/9 folds)
- CHAD: selection frequency=0.444 (4/9 folds)
- FOXA1: selection frequency=0.444 (4/9 folds)
- SLC44A4: selection frequency=0.333 (3/9 folds)
- STC2: selection frequency=0.333 (3/9 folds)
- BMPR1B: selection frequency=0.333 (3/9 folds)
- GP2: selection frequency=0.333 (3/9 folds)
- CHR1: selection frequency=0.333 (3/9 folds)
- STMND1: selection frequency=0.333 (3/9 folds)
- CST9: selection frequency=0.222 (2/9 folds)
- CPB1: selection frequency=0.222 (2/9 folds)
- SYT9: selection frequency=0.222 (2/9 folds)

- PRR15: selection frequency=0.111 (1/9 folds)
- EVL: selection frequency=0.111 (1/9 folds)
- WNK4: selection frequency=0.111 (1/9 folds)
- IL20: selection frequency=0.111 (1/9 folds)

Feature-screening comparison note: only methods actually executed in this Analysis are reported; unexecuted screening algorithms are not presented as comparative evidence.

4. Modeling · Methods and Comparative Performance

Candidate genes entering Modeling: 50; selected-model signature genes: 30

- Selected · Logistic Regression: algorithm=model.logistic_regression v1.0.0; repeated-CV AUC=0.964 ± 0.015; CV accuracy=0.929; selected genes=30; fold Jaccard=0.447; selection=embedded_l1_ranking_then_l2_logistic_fit

Model-specific selected genes: ESR1, TBC1D9, FOXA1, PGR, DACH1, CYP2B7P, POTEKP, SLC44A4, CA12, DNALI1, LINC00993, RGS22, ABAT, GATA3, SLC7A2, AGR2, ERBB4, C5AR2, AGR3, DNAJC12, DEGS2, GATA3-AS1, SLC16A6, CT62, TNRC18P1, ADAMTS15, SLC39A6, KCNJ3, GREB1, TFF3

Top fold-stable genes: ESR1 (1.00), TBC1D9 (1.00), PGR (1.00), CYP2B7P (1.00), DACH1 (1.00), AGR3 (0.89), DNALI1 (0.89), POTEKP (0.89), LINC00993 (0.89), CA12 (0.89), TFF1 (0.78), TNRC18P1 (0.78), AGR2 (0.78), SLC7A2 (0.78), GATA3 (0.78), KCNJ3 (0.67), RGS22 (0.67), ABAT (0.67), DNAJC12 (0.67), DEGS2 (0.67), SLC16A6 (0.67), KCNK15 (0.67), AFF3 (0.67), GATA3-AS1 (0.67), NAT1 (0.56), ZMYND10 (0.56), FSIP1 (0.44), ADAMTS15 (0.44), GREB1 (0.44), SERPINA11 (0.44), CLSTN2 (0.44), SLC44A4 (0.33), C5AR2 (0.33), BMPR1B (0.33), FOXA1 (0.33), STC2 (0.33), CHAD (0.33), SLC39A6 (0.33), THSD4 (0.33), ERBB4 (0.33), CHRD (0.33), CT62 (0.33), MLPH (0.33), MAPT (0.22), GFRA1 (0.22), TPRG1 (0.22), CCDC170 (0.22), CST9 (0.22), AKR7A3 (0.22), GP2 (0.22), CPB1 (0.22), KCNE4 (0.22), SUSP3 (0.11), SCUBE2 (0.11), LINC01087 (0.11), ANKRD30A (0.11), WNK4 (0.11), PSAT1 (0.11), STMND1 (0.11), IL20 (0.11)

Apparent outer-training fit: ROC AUC=0.9892; PR-AUC=0.9969; accuracy=0.9518; balanced accuracy=0.9564; sensitivity=0.9479; specificity=0.9648; reference threshold=0.50; training N=622. Apparent/resubstitution only; not validation evidence.

Repeated-CV OOF diagnostics: ROC AUC=0.9650; PR-AUC=0.9872; accuracy=0.9389; balanced accuracy=0.9331; sensitivity=0.9437; specificity=0.9225; reference threshold=0.50; OOF N=622.

- Comparison · Random Forest: algorithm=model.random_forest v1.0.0; repeated-CV AUC=0.953 ± 0.021; CV accuracy=0.943; selected genes=30; fold Jaccard=0.565; selection=random_forest_candidate_ranking_then_refit

Model-specific selected genes: ESR1, AGR3, CCDC170, GATA3, C5AR2, TBC1D9, CT62, POTEKP, THSD4, DNALI1, LINC01087, DEGS2, NAT1, ZMYND10, CYP2B7P, CA12, SLC39A6, DACH1, MLPH, GATA3-AS1, MAPT, SERPINA11, PGR, RGS22, SLC16A6, PSAT1, AGR2, ABAT, TFF1, SLC7A2

Top fold-stable genes: ESR1 (1.00), CCDC170 (1.00), AGR3 (1.00), GATA3 (1.00), C5AR2 (1.00), POTEKP (1.00), FSIP1 (1.00), SLC39A6 (1.00), TBC1D9 (1.00), THSD4 (1.00), CT62 (0.89), ZMYND10 (0.89), LINC01087 (0.89), GATA3-AS1 (0.89), CA12 (0.89), PSAT1 (0.89), DNALI1 (0.89), CYP2B7P (0.78), ABAT (0.78), NAT1 (0.78), DEGS2 (0.67), DACH1 (0.67), RGS22 (0.67), KCNK15 (0.67), MLPH (0.56), PGR (0.56), AGR2 (0.56), TFF1 (0.56), SERPINA11 (0.56), GREB1 (0.44), AFF3 (0.44), SCUBE2 (0.44), MAPT (0.44),

SLC44A4 (0.33), ERBB4 (0.33), SLC7A2 (0.33), GFRA1 (0.33), GRPR (0.33), CLSTN2 (0.33), KCNJ3 (0.22), STC2 (0.22), SLC16A6 (0.22), FOXA1 (0.22), ADAMTS15 (0.22), BMPR1B (0.22), GP2 (0.11), CHRD (0.11), TNRC18P1 (0.11), PRR15 (0.11), EVL (0.11), SUSP3 (0.11), TPRG1 (0.11), CHAD (0.11)

Apparent outer-training fit: ROC AUC=1.0000; PR-AUC=1.0000; accuracy=1.0000; balanced accuracy=1.0000; sensitivity=1.0000; specificity=1.0000; reference threshold=0.50; training N=622.
Apparent/resubstitution only; not validation evidence.

Repeated-CV OOF diagnostics: ROC AUC=0.9508; PR-AUC=0.9749; accuracy=0.9518; balanced accuracy=0.9291; sensitivity=0.9708; specificity=0.8873; reference threshold=0.50; OOF N=622.

Selected model rationale: Logistic Regression was chosen using repeated-CV performance before untouched-holdout evaluation.

4.1 Model Comparison, Concordance & Reproducibility

- Logistic Regression was selected from the compared algorithms using the prespecified repeated-CV selection rule.
- The selected model achieved an untouched-holdout ROC AUC of 0.970 and PR-AUC of 0.987.
- Cross-model concordance identified 20 genes shared across all successfully compared model-specific selected-gene sets.
- Biological convergence summarized 8 shared and 0 partially shared model-supported pathways.

Selected model: Logistic Regression (model.logistic_regression)

Modeling run ID: Omitted in portfolio version

Cross-model concordance uses outer-training repeated-CV evidence; the untouched holdout is not used for candidate-model comparison.

5. Heatmap Visualization

Displayed genes: SCGB2A2, SCGB1D2, CPB1, TFF1, PIP, CLEC3A, S100A7, AGR3, LINC00993, LTF, CALML5, TFF3

6. Signature Score

Signature genes: 25; method: delta-weighted signature score: 0 vs 1

Top signature genes: AGR3, ESR1, TFF1, CYP2B7P, NAT1, SCUBE2, POTEKP, FSIP1, CCDC170, GFRA1

- Signature scores summarize each sample using the selected biomarker genes and signed expression weights.
- Higher scores indicate stronger alignment with the positive/disease-like signature when two-class metadata is available.
- Use this panel to connect feature selection, modeling, heatmap visualization, and report-ready biomarker scoring.

7. Simulation Lab

Target: ESR1; mode: inhibition / down-regulation; mean delta=0.0000

- BioFlow simulated inhibition / down-regulation of ESR1 and recomputed a biomarker signature score.
- A positive delta indicates movement toward the current disease-like signature direction; a negative delta suggests movement away from it.
- This is an exploratory in silico sensitivity analysis, not experimental validation or a causal claim.

8. Validation & Clinical Utility

ROC AUC=0.970; PR-AUC=0.987; Brier=0.041

Positive class: 1; n=778; features=30

- ROC/AUC and PR-AUC are calculated from the exact out-of-sample holdout predictions created during Modeling.
- Validation does not refit the model or reselect features, preventing the validation holdout from leaking back into model training.
- Calibration, Brier score, threshold metrics, and decision-curve analysis are derived from those same saved holdout probabilities.
- Cross-validation is intentionally not recomputed here; repeated or nested CV belongs upstream in Modeling.
- Independent external validation remains necessary before making strong generalizability or clinical-utility claims.

8.1 Independent External Validation

External cohort: n=116; overlap features=30; positive class=Positive

External ROC AUC=0.967; External PR-AUC=0.996; External accuracy=0.784; External Brier=0.156

Sensitivity=0.757; Specificity=1.000

Reference/prespecified threshold=0.50; status=severe_shift; accuracy=0.784; sensitivity=0.757; specificity=1.000; confirmatory eligible=Yes.

Exploratory/user-adjusted threshold=0.10; status=stable; accuracy=0.931; sensitivity=0.922; specificity=1.000; confirmatory eligible=No. This is a reportable exploratory finding, not a replacement for the prespecified threshold.

8.2 External Multi-Model Benchmark

Locked Modeling winner=model.logistic_regression; winner external descriptive rank=1. Model selection remains the repeated-CV decision made before external outcomes were examined; external ranks are descriptive transportability observations only.

- Modeling winner · Logistic Regression: Internal CV rank=1; Internal CV AUC=0.964; External descriptive rank=1; External AUC=0.967; PR-AUC=0.996; Brier=0.156; Balanced accuracy=0.879; Reference threshold=0.50; Protocol=DIAGNOSTIC_ONLY; Portability=PARTIALLY_RETAINED.
- Frozen candidate · Random Forest: Internal CV rank=2; Internal CV AUC=0.953; External descriptive rank=2; External AUC=0.948; PR-AUC=0.994; Brier=0.070; Balanced accuracy=0.807; Reference threshold=0.50; Protocol=DIAGNOSTIC_ONLY; Portability=RETAINED.

The external benchmark never reselects or activates a candidate, retrain a model, or changes a reference threshold from external outcomes. A candidate outperforming the winner externally is reported only as descriptive algorithm-specific transportability divergence.

- Validation protocol: DIAGNOSTIC_ONLY. Cohort independence: pass.
- Feature compatibility: 30/30 required model features matched.
- Measurement compatibility after any registered conversion: incompatible; model-feature distribution shift: low.
- Representation conversion: condition_required; declared external $\log_2(\text{FPKM} + 0.1)$ -> effective model input $\log_2(\text{FPKM} + 0.1)$.
- Model portability: PARTIALLY_RETAINED. Reference/prespecified threshold portability at 0.50: severe_shift.
- Exploratory threshold 0.10 was also evaluated and is saved as exploratory/user-adjusted evidence only; it does not replace the confirmatory reference threshold 0.50.
- Discrimination, calibration, threshold performance, majority-class baseline, PR no-skill baseline, and Brier skill are reported separately.
- BioFlow did not use external outcome labels for representation conversion or cohort-distribution alignment.

9. AI Scientist Perturbation Reasoning

Target: ESR1; Disease: breast cancer; Direction: loss_of_function; Confidence: high

Target priority: 78/100 (high)

Evidence chain:

- ESR1 has a curated pathway entry in BioFlow knowledge base.
- ESR1 was found in the uploaded expression matrix as ESR1.
- Strongest positive expression neighbors: CCDC170 $r=0.87$, AGR3 $r=0.85$, THSD4 $r=0.85$, TBC1D9 $r=0.85$, C5AR2 $r=0.84$.
- Disease-module overlap detected: er pr luminal program (FOXA1, GATA3, XBP1).
- Candidate pathway mechanisms: ER/PR hormone-receptor signaling, luminal differentiation program, endocrine-therapy response biology.
- Candidate phenotype consequences: estrogen-dependent tumor growth, luminal breast cancer identity, endocrine sensitivity or resistance, estrogen-dependent growth.

AI interpretation:

AI Perturbation Simulation Summary

BioFlow evaluated ESR1 under a Inhibition scenario in the context of breast cancer using a lightweight Geneformer-inspired reasoning layer. A Inhibition perturbation is interpreted as reduced ESR1 activity. For this target, the first-order hypothesis is attenuation or disruption of ER/PR hormone-receptor signaling, with a downstream phenotype such as estrogen-dependent tumor growth. In breast cancer, prioritize ER/PR/HER2 subtype biology, PI3K-AKT/MAPK growth signaling, TP53/cell-cycle control, BRCA/DNA-repair vulnerability, EMT/invasion, and immune microenvironment endpoints.

Disease-context priority: Prioritize ER/PR/HER2 subtype biology, PI3K-AKT/MAPK growth signaling, TP53/cell-cycle checkpoints, BRCA/DNA repair vulnerability, EMT/invasion, angiogenesis/hypoxia, and immune microenvironment hypotheses.

For ESR1, BioFlow emphasizes ER-positive luminal breast cancer biology, endocrine-response signaling, and endocrine resistance hypotheses.

Candidate Mechanisms

The most relevant candidate pathway modules include: ER/PR hormone-receptor signaling, luminal differentiation program, endocrine-therapy response biology. Potential phenotype-level consequences include: estrogen-dependent tumor growth, luminal breast cancer identity, endocrine sensitivity or resistance, estrogen-dependent growth. These outputs should be interpreted as mechanistic hypotheses rather than experimentally confirmed effects.

Uploaded-data Context

The target gene was found in the currently uploaded expression table. Detected layout: genes_as_rows; gene axis: first_column. Matched feature: ESR1.

Expression-context Reasoning

Because some targets have limited curated annotation, BioFlow also performed data-driven co-expression inference across the uploaded samples. The strongest positively associated genes for ESR1 were: CCDC170 ($r=0.87$), AGR3 ($r=0.85$), THSD4 ($r=0.85$), TBC1D9 ($r=0.85$), C5AR2 ($r=0.84$), GATA3 ($r=0.84$), CA12 ($r=0.82$), XBP1 ($r=0.80$). Strong inverse associations included: PSAT1 ($r=-0.74$), VGLL1 ($r=-0.73$), PPP1R14C ($r=-0.71$), BCL11A ($r=-0.70$), B3GNT5 ($r=-0.68$).

Inferred breast cancer modules: er pr luminal program via FOXA1, GATA3, XBP1.

If ESR1 is inhibited or knocked down, the co-expressed module may be reduced, disconnected, or compensated, depending on whether ESR1 is causal or only a marker of that module. This is a guilt-by-association hypothesis, not proof of causality; it is useful for prioritizing enrichment analysis, feature ablation, and experimental validation.

Why BioFlow inferred this

- `curated_gene_prior` (high): ESR1 has a curated pathway entry in BioFlow knowledge base.
- `uploaded_data_match` (moderate): ESR1 was found in the uploaded expression matrix as ESR1.
- `coexpression` (moderate): Strongest positive expression neighbors: CCDC170 $r=0.87$, AGR3 $r=0.85$, THSD4 $r=0.85$, TBC1D9 $r=0.85$, C5AR2 $r=0.84$.
- `module_overlap` (moderate): Disease-module overlap detected: er pr luminal program (FOXA1, GATA3, XBP1).
- `pathway_hypothesis` (hypothesis): Candidate pathway mechanisms: ER/PR hormone-receptor signaling, luminal differentiation program, endocrine-therapy response biology.
- `phenotype_hypothesis` (hypothesis): Candidate phenotype consequences: estrogen-dependent tumor growth, luminal breast cancer identity, endocrine sensitivity or resistance, estrogen-dependent growth.
- `related_genes` (supporting): Related genes for validation: CCND1, FOXA1, GATA3, GREB1, PGR, TFF1, AR, XBP1.
- `curated_literature_prior` (high): ESR1 is a key luminal breast cancer transcriptional regulator and endocrine-response context marker.

Predicted cell-state consequences

- General disease-relevant expression module: requires validation by enrichment or cell-state scoring; evidence: weak or exploratory module evidence

Drug-like perturbation view

This resembles a endocrine-therapy-like perturbation. Candidate modality examples: estrogen receptor blockade, endocrine therapy class. Rationale: reduced ER-driven transcriptional and proliferative programs.

PubMed-style evidence panel

- `curated_literature_prior`: ESR1 is a key luminal breast cancer transcriptional regulator and endocrine-response context marker.

- `curated_literature_prior`: ESR1 perturbation hypotheses should be validated with PGR, FOXA1, GATA3, GREB1, and proliferation/endocrine-resistance signals.
- `therapeutic_prior`: endocrine-therapy-like perturbation suggests a translational intervention framing: reduced ER-driven transcriptional and proliferative programs

Enrichment analysis

BioFlow performed local hypergeometric enrichment on the target co-expression neighborhood. Top enriched modules: `er pr luminal program` (FDR=1.06e-07, overlap=3/25)

Disease-state comparison

Using metadata label `ER_status`, BioFlow compared ESR1 expression between Positive (n=600) and Negative (n=178). Mean difference `group_b - group_a` = -5.3; Welch p-value = 4.480861870891273e-108. Direction: `higher_in_Positive`.

Target prioritization

BioFlow target priority is high with overall score 0.78. Component scores: pathway relevance=0.8, expression context=0.75, enrichment support=0.75, disease-state signal=0.85, drug-like actionability=0.75.

Confidence evidence scoring

Pathway evidence=0.85; expression evidence=0.75; module evidence=0.5; literature-style prior=0.8. Overall confidence is high (0.73).

Suggested Validation

Recommended next steps are: differential expression or gene-set enrichment after perturbation, single-cell state transition analysis if scRNA-seq is available, feature ablation/permutation tests in the trained ML model, and independent experimental or literature validation. Confidence for this *in silico* hypothesis is high.

Interpretation note: Independent external validation is included for the recorded external cohort. Broader transportability and mechanistic or clinical claims still require evidence appropriate to those claims.